FORMALIZING ANGULAR STRUCTURE OF PARTICLE JETS

By

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ABSTRACT OF THE DISSERTATION

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Acknowledgments

A nice thank you...

Dedication

A dedication here.

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Chapter 1

Introduction

1.1 The Standard Model

According to the Standard Model (SM), all matter results from interactions between fundamental particles: lepton and quarks (fermions), and mediators (bosons). Table 1.1 shows fermions of the standard model, their anti-particle equivalent would have all the signs reversed for the quantum numbers. For example, an anti-electron (e^+) and anti-electron-neutrino ($\bar{\nu_e}$) lepton number would flip to $(-1, 0, 0)$ and charges (Qs) would be $+1$ and 0 respectively.

	Leptons			Quarks		
	l	Q	(L_e, L_μ, L_τ)	q	Q	$(F_d, F_u, F_s, F_c, F_b, F_t)$
First generation	e^-	-1	(1, 0, 0)	d	$-1/3$	$(-1, 0, 0, 0, 0, 0)$
	ν_e	0	(1, 0, 0)	u	$2/3$	$(0, 1, 0, 0, 0, 0)$
Second generation	μ^-	-1	(0, 1, 0)	s	$-1/3$	$(0, 0, -1, 0, 0, 0)$
	ν_μ	0	(0, 1, 0)	c	$2/3$	$(0, 0, 0, 1, 0, 0)$
Third generation	τ^-	-1	(0, 0, 1)	b	$-1/3$	$(0, 0, 0, 0, -1, 0)$
	ν_τ	0	(0, 0, 1)	t	$2/3$	$(0, 0, 0, 0, 0, 1)$

Table 1.1: Three generations of leptons (left) and quarks (right).

The mediators/bosons exchanged between the other fundamental particles give rise to the fundamental interactions: electromagnetism from the photon (γ), weak force from $W^\pm$ and Z bosons, the gluon (g) for the strong force, and the Higgs boson (h) for imparting mass to the fundamental particles. Various interaction vertices possible in SM, that become the building blocks of all fundamental interactions are shown in 1.2.

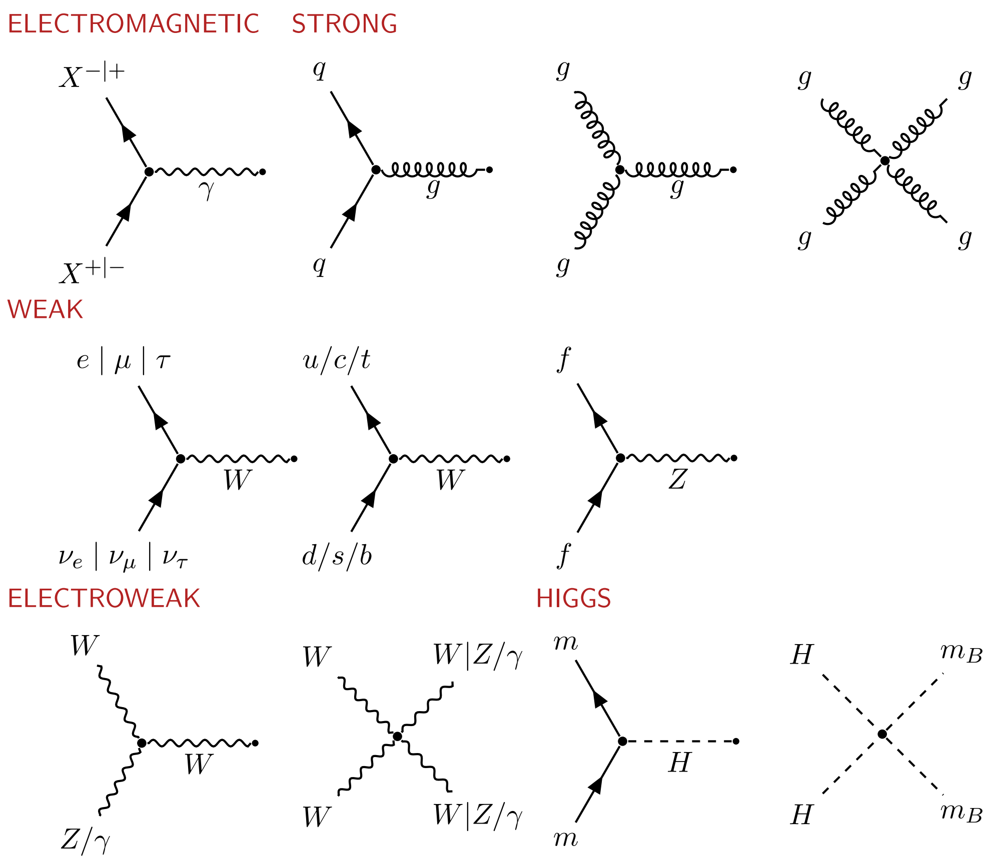


Table 1.2: Feynman diagrams of various interaction vertices possible in the Standard Model

1.2 Quantum Chromodynamics

1.2.1 Lagrangian and Feynman Rules

Quantum Chromodynamics (QCD) is the component of the SM which describes the strong interactions of quarks and gluons. The QCD Lagrangian can be expected as:

$$\mathcal{L}_{\mathrm{QCD}} = \underbrace{\bar{q}^{\alpha}(i\gamma^{\mu}\partial_{\mu}\delta_{\alpha\beta} - g\gamma^{\mu}t^{a}_{\alpha\beta}A^{a}_{\mu} - m\delta_{\alpha\beta})q^{\beta} - \frac{1}{4}F^{a}_{\mu\nu}F^{\mu\nu}_{a}}_{\mathcal{L}_{\mathrm{classical}}}$$
$$\underbrace{-\frac{1}{2\lambda}(\partial^{\mu}A^{a}_{\mu})^{2}}_{\mathcal{L}_{\mathrm{gauge-fixing}}} + \underbrace{\bar{c}^{\alpha}\partial^{\mu}(\partial_{\mu}\delta_{\alpha\beta} + igT^{a}_{\alpha\beta}A^{a}_{\mu})c^{\beta}}_{\mathcal{L}_{\mathrm{ghost}}} \tag{1.1}$$

$\mathcal{L}_{\mathrm{classical}}$ is the classical Lagrangian density shows interaction of a quark field q^{α} with 3 color degrees of freedom interacting with gluon fields A^{a}_{μ} with 8 color degrees of freedom. g and m are dimensionless and correspond to QCD coupling constant, and quark-mass respectively. $F^{a}_{\mu\nu}$ are the gluon tensors defined as,

$$F^{a}_{\mu\nu} = \partial_{\mu}A^{a}_{\nu} - \partial_{\nu}A^{a}_{\mu} - gf_{abc}A^{b}_{\mu}A^{c}_{\nu}. \tag{1.2}$$

To properly define gluon propagators, we need to make a choice of gauge. The $\mathcal{L}_{\mathrm{gauge-fixing}}$ parametrizes the class of covariant gauges with the gauge parameter λ. In non-Abelian theories, this gauge fixing term leads to creation of unphysical degrees of freedom, which are canceled out by $\mathcal{L}_{\mathrm{ghost}}$ which contains complex, scalar fermion fields c^{α}.

The matrices t^{a} and T^{a} are the fundamental and adjoint representations of SU(3) in form of traceless 3×3 and 8×8 hermitian matrices satisfying:

$$[t^{a}, t^{b}] = if_{abc}t^{c},\ \mathrm{tr}\Big(t^{a}t^{b}\Big) = \frac{1}{2}\delta^{ab}$$
$$[T^{a}, T^{b}] = if_{abc}T^{c},\ (T^{a})_{bc} = -if^{abc}. \tag{1.3}$$

Comparing with quantum electrodynamics (QED), where the photon field tensor lacks the last term of Eqn. 1.2 as it is an Abelian, $U(1)$ gauge theory, we can explain the presence of three/four-gluon vertices and absence of such vertices for photon in Table 1.2.

1.2.2 The Running Coupling

At an energy scale Q, consider a dimensionless observable X. To calculate X as a perturbation series in the QCD coupling $\alpha_s = \frac{g^2}{4\pi}$ (analogous to the QED fine structure constant), we need to remove ultraviolet divergences through renormalization. Renormalization introduces another arbitrary energy scale μ where the divergences are subtracted. In general, X would depend on the ratio Q^2/μ^2 and post-renormalization $\alpha_s(\mu^2)$. However, any physical observable should be independent of μ:

$$\begin{aligned} \mu^2 \frac{\partial X(Q^2/\mu^2, \alpha_s(\mu^2))}{\partial \mu^2} &= \left[\mu^2 \frac{\partial}{\partial \mu^2} + \mu^2 \frac{\partial \alpha_s(\mu^2)}{\partial \mu^2} \frac{\partial}{\partial \alpha_s}\right] X(Q^2/\mu^2, \alpha_s(\mu^2)) = 0 \\ &\implies \left[-\frac{\partial}{\partial t} + \beta(\alpha_s) \frac{\partial}{\partial \alpha_s}\right] X(e^t, \alpha_s) = 0. \end{aligned} \tag{1.4}$$

Where,

$$\begin{aligned} & t = \ln \frac{Q^2}{\mu^2}, \quad \beta(\alpha_s) = \mu^2 \frac{\partial \alpha_s}{\partial \mu^2}, \quad \alpha_s \equiv \alpha_s(\mu^2(t)) \\ &\implies \beta(\alpha_s) = -\frac{\partial \alpha_s}{\partial t} \\ &\implies t - t_0 = -\int_{\alpha_s(\mu^2(t_0))}^{\alpha_s(\mu^2(t))} \frac{d\alpha_s}{\beta(\alpha_s)} \end{aligned} \tag{1.5}$$

Now, setting $t_0 = 0$ gives $\mu^2(t_0) = Q^2$ and the definite integral becomes:

$$t = \int_{\alpha_s}^{\alpha_s(Q^2)} \frac{d\alpha_s}{\beta(\alpha_s)} \tag{1.6}$$

Substituting in Eq. 1.4:

$$\frac{\partial \alpha_s(Q^2)}{\partial t} = \beta(\alpha_s(Q^2)), \quad \frac{\partial \alpha_s(Q^2)}{\partial \alpha_s} = \frac{\beta(\alpha_s(Q^2))}{\beta(\alpha_s)} \tag{1.7}$$

we see that $X(1, \alpha_s(Q^2))$ is a valid solution. Thus the scale dependence in X comes from the *running coupling* $\alpha_s(Q^2)$.

Taking derivative of Eq. 1.6 with respect to Q^2 gives the *renormalization group equation*,

$$Q^2 \frac{\partial \alpha_s}{\partial Q^2} = \beta(\alpha_s) \equiv -\alpha_s^2(Q^2) \left(\beta_0 + \beta_1 \alpha_s(Q^2) + \mathcal{O}(\alpha_s^2(Q^2))\right) \tag{1.8}$$

Where,

$$\beta_0 = \frac{1}{(4\pi)^2}\left(11 - \frac{2}{3}N_f\right), \quad \beta_1 = \frac{1}{(4\pi)^4}\left(102 - \frac{38}{3}N_f\right) \tag{1.9}$$

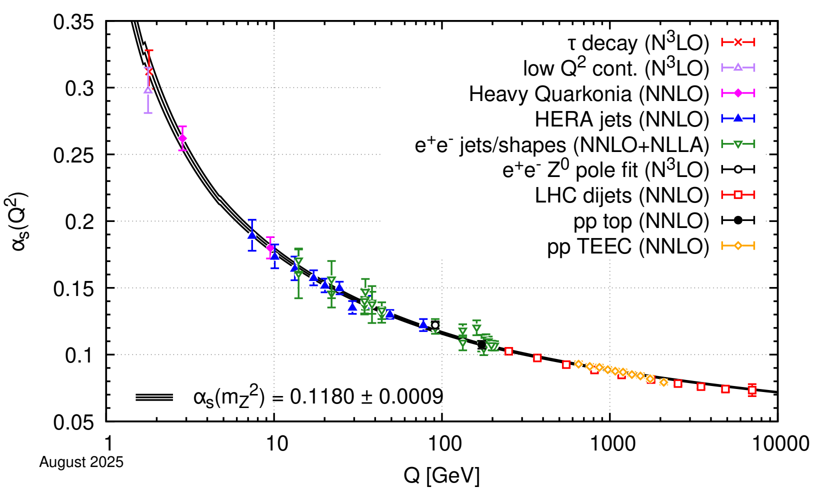


Figure 1.1: Summary of determinations of α_s as a function of the energy scale Q compared to the running of the coupling computed at five loops taking as an input the current Particle Data Group (PDG) world average, $\alpha_s(m_Z^2) = 0.1180 \pm 0.0009$ [1]

Integrating Eq. 1.8

$$\begin{aligned} \ln\left(\frac{Q^2}{Q_0^2}\right) &= \int_{\alpha_s(Q_0^2)}^{\alpha_s(Q^2)} \frac{d\alpha_s}{\beta(\alpha_s)} \\ \implies \alpha_s(Q^2) &= \frac{\alpha_s(Q_0^2)}{1+\beta_0(\alpha_s(Q_0^2))\ln\left(Q^2/Q_0^2\right)+\mathcal{O}(\alpha_s^2)} \end{aligned} \tag{1.10}$$

Usually, the reference scale Q_0^2 is usually set to a convenient energy scale in the perturbative regime. α_s at all other scales can be inferred accordingly. Usually, Q_0 is set to M_Z as shown in Fig. 1.1. Another approach is to choose a *scale parameter* Λ such that $\alpha_s(\Lambda) \to \infty$.

$$\begin{aligned} \ln\left(\frac{Q^2}{\Lambda^2}\right) &= -\int_{\alpha_s(Q^2)}^{\infty} \frac{d\alpha_s}{\beta(\alpha_s)} \\ \implies \alpha_s(Q^2) &= \frac{1}{\beta_0 \ln(Q^2/\Lambda^2)}\left[1-\frac{\beta_1}{\beta_0}\frac{\ln\left(\ln\left(Q^2/\Lambda^2\right)\right)}{\ln(Q^2/\Lambda^2)}+\mathcal{O}\left(\frac{1}{\ln(Q^2/\Lambda^2)^2}\right)\right] \end{aligned} \tag{1.11}$$

Typically, $\Lambda \sim 200$MeV with small variations depending on its exact definition. This behaviour leads to:

- **Asymptotic Freedom:** At high momentum transfers ($Q \gg \Lambda$), the coupling strength vanishes ($\alpha_s \to 0$). In this regime, quarks and gluons behave as quasi-free particles, making perturbative QCD calculations highly reliable.

- **Color Confinement:** At low energy scales ($Q \sim \Lambda$, the coupling diverges ($\alpha_s \sim \mathcal{O}(1)$). In this strongly-interacting, non-perturbative limit, isolated color charges cannot exist; quarks and gluons are permanently bound into color-neutral hadrons.

1.3 QCD Jets

When a quark undergoes hard scattering during initial stages of a hadron-hadron collision , the interaction occurs at a scale where α_s is small. In this asymptotic regime, the probability

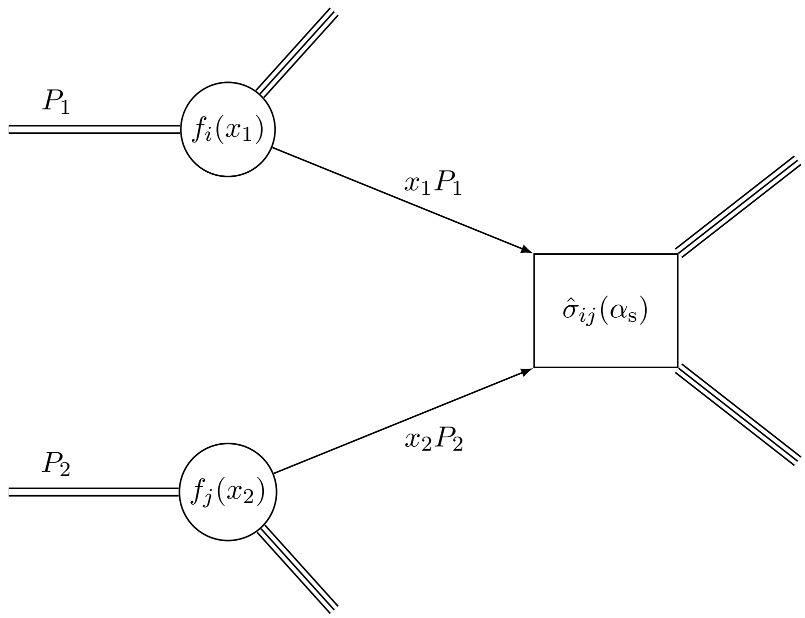


Figure 1.2: Collinear factorization of a hadronic hard scattering.

of the quark radiating a high-energy gluon is suppressed according to perturbative QCD, allowing the quark to initially behave as a free particle. However, as the energetic quark propagates outward, the characteristic momentum scale of its subsequent interactions begins to drop and $\alpha_s(\kappa^2)$ grows, which drastically increases the interaction probability at the fundamental vertices. The initial quark rapidly radiates gluons, which in turn split into more gluons and quark-antiquark pairs. This cascading sequence of emissions continues until the energy scale of the individual partons lowers to the confinement threshold, $\kappa \sim \Lambda_{\mathrm{QCD}}$, and the partons undergo hadronization. Experimentally this appears as a collimated spray of particles known as a *jet*.

1.4 Quark-Gluon Plasma

Chapter 2

Relativistic collisions of nuclei

2.1 Relativistic Kinematics

2.1.1 Laboratory vs Center-of-mass frames

A typical collision is a two-to-many reaction like:

$$\underbrace{A}_{\text{projectile}} + \underbrace{B}_{\text{target}} \to C_1 + C_2 + C_3 + \ldots, \tag{2.1}$$

In *laboratory frame*, the target is at rest, so,

$$\vec{p_B^{\text{lab}}} = \vec{0} \implies p_B^{\text{lab}} = (m_B, 0, 0, 0), \tag{2.2}$$

In *center-of-mass* frame, the sum of three-momentum vectors vanish in the initial and final states

$$\vec{p_A^{\text{cm}}} + \vec{p_B^{\text{cm}}} = \sum_i \vec{p_{C,i}^{\text{cm}}} = \vec{0}, \tag{2.3}$$

Using Lorentz invariance of the Mandelstam variable $s \equiv (p_A + p_B)_\mu (p_A + p_B)^\mu$:

$$\begin{aligned} s &= (E_A^{\text{lab}} + m_B)^2 - |\vec{p_A^{\text{lab}}}|^2 = (E_A^{\text{cm}} + E_B^{\text{cm}})^2 \\ \implies s &= m_A^2 + m_B^2 + 2m_B E_A^{\text{lab}} = (E^{\text{cm}})^2 \end{aligned} \tag{2.4}$$

As relativistic energies ($\mathcal{O}(10^{2-3}\text{GeV})$) are much higher than reactant masses $\mathcal{O}(1 - 10\text{GeV})$,

$$\sqrt{s} = E^{\mathrm{cm}} \approx \sqrt{2 m_B E_A^{\mathrm{lab}}} \tag{2.5}$$

As the center-of-mass energy that produces new particles increases only as $\sqrt{E^{\mathrm{lab}}}$, making any gain in particle production more expensive at relativistic energies. Thus modern colliders usually operate in center-of-mass frames.

2.1.2 Coordinate system

Considering the non-linear addition of velocities in relativistic mechanics,

$$\beta = \frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2}, \tag{2.6}$$

Considering the transformation $y = \tanh^{-1} \beta$,

$$\begin{aligned} \tanh^{-1} \beta_1 + \tanh^{-1} \beta_2 = \tanh^{-1} \frac{\beta_1 + \beta_2}{1 + \beta_1 \beta_2} = \tanh^{-1} \beta \\ \implies y_1 + y_2 = y \end{aligned} \tag{2.7}$$

The addition is linear again. We call this transformed velocity *rapidity*, given by:

$$y = \tanh^{-1} \beta = \frac{1}{2} \ln \frac{1 + \beta}{1 - \beta} = \frac{1}{2} \ln \frac{E + p_z}{E - p_z} \tag{2.8}$$

Rapidity can grow without bound, and at small velocities, $y \approx \beta$, making it a relativistic analogue of velocity. Since it transforms linearly under Lorentz boosts, rapidity distribution of particles retain their shapes.

For relativistic energies, $E \approx p$, making 2.8,

$$y \approx \frac{1}{2} \ln \frac{p + p_z}{p - p_z} = -\ln \left(\tan \frac{\theta}{2} \right) = \eta \tag{2.9}$$

η is called the *pseudo-rapidity*, the massless limit of rapidity. As shown in 2.1.2, pseudo-rapidity can be directly determined form particle production angle θ measured with respect to the beam axis.

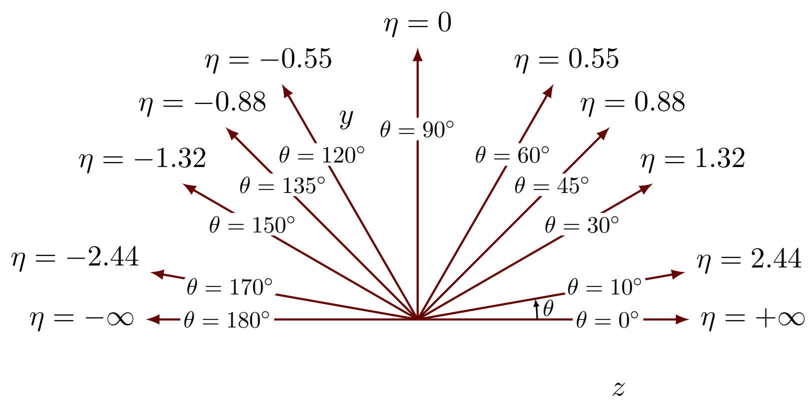


Figure 2.1: η vs θ

Also considering *azimuthal angle* (φ) and *transverse-momentum* $\left(p_{\mathrm{T}} = \sqrt{p_x^2 + p_y^2}\right)$ of the particles, which are invariant under boosts along the beam-line by design, we can use $(p_{\mathrm{T}}, \eta, \varphi)$ as the coordinate system for the particles produced in the collision. Figure 2.2 illustrates this coordinate system for a cylindrical collider detector.

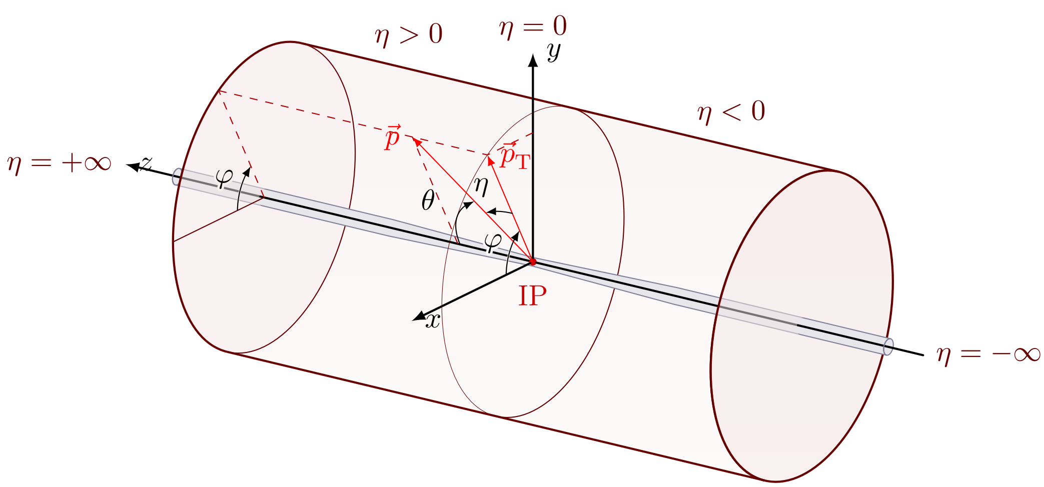


Figure 2.2: The $(p_{\mathrm{T}}, \eta, \varphi)$ coordinate system, drawn for a cylindrical collider detector. The beam axis defines z and the interaction point (IP) sits at the origin. A particle of momentum $\vec{p}$ is described by its transverse momentum $\vec{p}_{\mathrm{T}}$, its azimuthal angle φ in the transverse plane, and its pseudo-rapidity η, fixed by the polar angle θ with respect to the beam.

2.2 Heavy-Ion Collisions

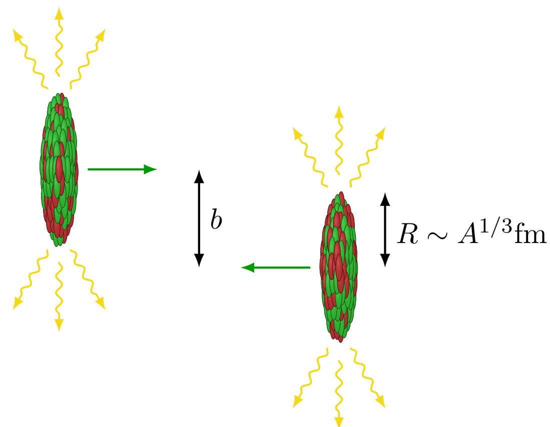
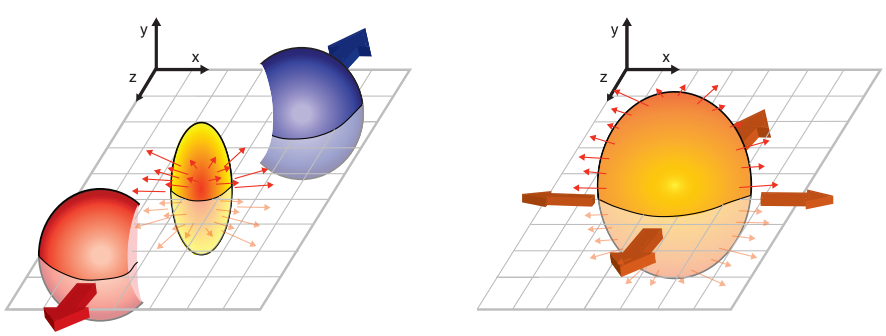


Figure 2.3: Typical heavy-ion collision

2.2.1 Centrality determination and event plane reconstruction

Centrality is a measure of the transverse overlap between the colliding nuclei and is generally expressed as a percentage of all collisions. For example, the 0-10% most central events would refer to the 10% of events with the most overlap and thus the 10% smallest impact parameter. This analysis studied semi-peripheral (20-50%) events to maximize the eccentricity of the interaction region. Centrality is determined by fitting the charged-particle multiplicity from the TPC within $|\eta| < 0.5$ that is corrected for dependence on the v_z and the beam luminosity.

Within the overlap region, symmetry planes are generated from initial asymmetries in the nucleon distributions and can be quantified by a harmonic decomposition [3]. The reaction plane would correspond to the second-order symmetry plane $\Psi_{2,EP}$ if nucleon distributions



were in their average positions and devoid of fluctuations of interactions amongst nucleons [4]. We refer in this letter to the event plane as being the experimentally reconstructed second-order symmetry plane [3].

By measuring the charged particle azimuthal distribution, the n-th order event plane can be extracted by [3]:

$$\Psi_{n,EP} = \frac{1}{n} \arctan\left(\frac{Q_{y,n}}{Q_{x,n}}\right), \tag{2.10}$$

Where, the weighted Q vectors are given by:

$$Q_{x,n} = \sum_{\text{tracks}} w_{\text{track}} \cos(n\phi_{\text{track}}) \qquad \& \qquad Q_{y,n} = \sum_{\text{tracks}} w_{\text{track}} \sin(n\phi_{\text{track}}). \tag{2.11}$$

where the sum is calculated for all reconstructed charged particles (tracks) in the event, ϕ_{track} is the track's azimuthal angle, and w_{track} the weight associated with the track. Weights, w_{track}, are optimized to calculate the event-plane vector to the best accuracy. This work uses the common approach of scaling by the track's p_{T} ($w_{\text{track}} = p_{\text{T,track}}$) [3]. The event plane is calculated event-by-event, following a procedure similar to Ref. [5] using charged tracks with $0.2 < p_{\text{T,track}} < 1.0$ GeV/c measured within the TPC. The approach is called the Modified Reaction Plane (MRP) method [2]. Additional details can be found in Refs. [5, 6].

The impact of highly energetic jets on the calculation of the event-plane orientation is reduced by removing the particles within the pseudorapidity strip ($|\Delta\eta| < 0.4$) across $\Delta\phi$ surrounding the leading jet. This procedure also removes a significant portion of the away-side jet, located opposite in azimuth. An upper limit of 1.0 GeV/c is used in the calculation of the event plane to exclude the momentum range of particles used in correlation functions from the calculation of the event plane which is used to characterize the near-side jets. Due to finite acceptance and multiplicities, the calculated event plane has an underlying anisotropy that is corrected by applying two separate correction methods.

First, a calibration and recentering correction procedures are applied to remove bias introduced by non-uniform acceptance of the TPC tracking system and further account for potential beam-condition effects[3, 7, 8]. This procedure involves recentering the weighted Q-vectors such that, $\langle Q_{x,n}\rangle = 0 = \langle Q_{y,n}\rangle$.

Recentering is done by calculating a modified Q-vector, obtained by subtracting an event averaged Q-vector from each event's nominal Q-vector and done for 10% centrality intervals and 4 cm v_z intervals. The recentering approach, which drastically improves the uniformity of the event plane, is however, unable to remove the higher harmonics of $\Psi_{n,EP}$ [3]. To help remove higher harmonics and make the event-plane angle isotropic in the lab frame [9], a second correction step, referred to as shifting, is applied event-by-event. This method defines a new angle:

$$\Psi'_{2,EP} = \Psi_{2,EP} + \sum_{n} \frac{2}{n}(-\langle \sin(n\Psi_{2,EP})\rangle \cos(n\Psi_{2,EP}) + \langle \cos(n\Psi_{2,EP}) \sin(n\Psi_{2,EP})\rangle), \quad (2.12)$$

where the brackets denote an average over events. We require the vanishing of each n-th Fourier moment up to 20th order. Similarly to recentering, the shifting correction is done separately for 10% centrality intervals and 4-cm v_z intervals. Additional details of the recentering and shifting corrections can be found in Refs. [3, 7, 10]

The resulting azimuthal anistropy can be characterized by the Fourier decomposition of the azimuthal particle distribution with respect to the second-order event plane [11, 12]:

$$\frac{dN}{d(\phi - \Psi_{RP})} = \frac{N_0}{2\pi}\left(1 + 2\sum_{n=1}^{\infty} v_n \cos[n(\phi - \Psi_{RP})]\right), \quad (2.13)$$

where N_0 is the number of particles, ϕ describes the azimuthal angle of the particles, Ψ_{RP} describes the azimuthal angle of the true reaction plane determined by the beam axis and the impact parameter and v_n is the n-th harmonic (flow) coefficient. Ψ_{RP} is not experimentally known and is replaced by the reconstructed event-plane angle. Due to finite event multiplicity, there will be a difference between these two planes. It is quantified by event-plane resolution, R_n or $R_n\{\Psi_{2,EP}\}$ given by Eqn. 2.14. The observed v_n, v_n^{obs} is corrected for this limited resolution by doing, $v_n = v_n^{\text{obs}}/R_n$ [3, 13]. Because an ideal event-plane resolution is equal to 1, for non-ideal cases, the value of the coefficients will be raised by applying the correction. Thus, R_n impacts the flow-modulated background for these correlations, as described in Sect. 4.1.3,

$$R_n = R_n\{\Psi_{2,EP}\} = \langle \cos(n[\Psi_{RP} - \Psi_{2,EP}]) \rangle \qquad n \text{ is even} \tag{2.14}$$

Furthermore, individual events are divided into two random sub-events by assigning charged tracks to sub-events, "a" and "b". The sub-events are unique, with approximately equal multiplicities. We can write the correlation of two event planes by taking the product of two sub-events [3, 14]:

$$\langle \cos\Big(n[\Psi^a_{2,EP} - \Psi^b_{2,EP}]\Big) \rangle = \langle \cos\big(n[\Psi^a_{2,EP} - \Psi_{RP}]\big) \rangle \langle \cos\Big(n[\Psi^b_{2,EP} - \Psi_{RP}]\Big) \rangle, \tag{2.15}$$

This allows calculation of the event-plane resolution directly from data. Since a and b have equal multiplicities, the resolution of each sub-event can be calculated from the correlation between the two sub-events as [3]:

$$\langle \cos\big(n[\Psi^a_{2,EP} - \Psi_{RP}]\big) \rangle = \sqrt{\langle \cos\Big(n[\Psi^a_{2,EP} - \Psi^b_{2,EP}]\Big) \rangle}. \tag{2.16}$$

The event-plane resolution is multiplicity dependent and calculated for separate ranges of collision centrality using the two sub-events method. Narrower bins are calculated and then combined accordingly to match the ranges used by this analysis, by averaging the results from the narrow bins weighted by the multiplicity of each bin [9].

2.3 Signatures of QGP

Different seminal results go here...

Chapter 3

Jets and Machine Learning

Chapter 4

Jet-hadron correlations relative to the event plane

Historically, the effects of the medium on high energy partons were studied by measuring the leading particles and studying the p_{T} distribution of pairs of high-p_{T} particles. Leading particle studies, however, have limitations in extreme quenching scenarios. In these cases, particle emission occurs near the surface of the medium and thus the sensitivity of the leading particles to the region of highest energy density is very limited. This consequently makes the nuclear modification factor insensitive to medium properties. Full jet reconstruction allows for unbiased measurement of the original parton 4-momentum and the transverse and longitudinal shape of the jet. Through modifications of the jet structure, properties of the medium can be studied.

When studying the $\Delta\phi$ distributions, a peak is observed at $\Delta\phi = 0$ which is the triggered jet. This side is referred to as the near-side. There is another peak which is seen near π. This is referred to as the away-side. The away-side consists of the hadrons associated with the trigger jet and make up the second jet which results from the early hard-scatter.

In this analysis, the coordinate system used to depict the distribution of associated particles is defined relative to a reconstructed jet, also called a trigger jet. The distribution is thus given by:

$$\frac{1}{N_{\mathrm{trig}}}\frac{d^2N_{\mathrm{assoc,jet}}}{d\Delta\phi d\Delta\eta}, \tag{4.1}$$

where N_{trig} is the number of trigger jets, $N_{\mathrm{assoc,jet}}$ is the number of associated particles, $\Delta\phi$ $(= |\phi_{\mathrm{jet}} - \phi_{\mathrm{assoc}}|)$ is the azimuthal angle of those trigger particles relative to the trigger jets, and $\Delta\eta$ ($= |\eta_{\rm jet} - \eta_{\rm assoc}|$) is the difference in the pseudorapidities of the trigger jet and associated particle. The goal of this analysis is to study the conditional yield of associated particles, the width of the near- and away-side peaks (quantified using the gaussian width) as a function of the angle between the jet axis and the event plane. The yield is estimated by:

$$\text{Yield} = \frac{1}{N_{\rm trig}} \int_c^d \int_a^b \frac{d^2 N_{\rm assoc,jet}}{d\Delta\phi d\Delta\eta} d\Delta\phi d\Delta\eta. \tag{4.2}$$

The choice of integration limits is somewhat arbitrary. They are chosen based on practical considerations, including the detector acceptance and binning of histograms.

4.1 Analysis Method

This measurement utilizes data collected during the 2014 run from Au+Au collisions at nucleon-nucleon center-of-mass energy of $\sqrt{s_{\rm NN}} ==$ 200 GeV by the STAR experiment [15] at RHIC. Events referred to as signal (same) events are required to contain a HT trigger in the BEMC [16]. Minimum-bias (MB) triggered events based on coincidence of Zero Degree Calorimeters (ZDC coincidence), Vertex Position Detectors and Beam-Beam Counters signals are used to estimate the pair-acceptance effects via a mixed-event (ME) technique [17]. For this analysis, 9.4M HT-triggered and 4.0M MB collision events are used. Events are further categorized by their centrality selection, defined in section 2.2.1. The events are required to have a reconstructed primary vertex $|v_z| < 24$ cm and centrality of 20-50%.

The reaction plane is approximated by the second-order event plane, which is the experimentally reconstructed second-order symmetry plane, and will be referred to as the "event plane" ($\Psi_{2,EP}$) in this text, for simplicity. The distributions of these associated tracks relative to the trigger jet are measured in three bins in the angle between the trigger jet and the event plane, in-plane ($|\Psi_{2,EP} - \phi_{\rm jet}| < \pi/6$), mid-plane ($\pi/6 < |\Psi_{2,EP} - \phi_{\rm jet}| < \pi/3$), and out-of-plane ($|\Psi_{2,EP} - \phi_{\rm jet}| > \pi/3$) bins. The analysis is restricted to 20–50% central

Au+Au collisions to achieve the highest event-plane resolution and therefore the analysis will be most sensitive to any path-length dependencies.

The second- (fourth-) order event-plane resolutions relative to the second-order event plane ($R_2\{\Psi_{2,EP}\}$ and $R_4\{\Psi_{2,EP}\}$ respectively) as a function of collision centrality are shown in Fig. 4.1. The resolution is peaked around the 20-30% and 30-40% centrality. The event-plane resolutions for $R_2\{\Psi_{2,EP}\}$ and $R_4\{\Psi_{2,EP}\}$ were 0.56 and 0.28, respectively. The errors on the event-plane resolution calculation were less than 1%, leading to a negligible effect on the final $\Delta\phi$ correlations. Measured values for the event-plane resolution are in good agreement with prior STAR studies [2]. These resolutions are evaluated to correct the observed flow coefficients which arise in the fits of the combinatorial background discussed in Sect. 4.1.3.

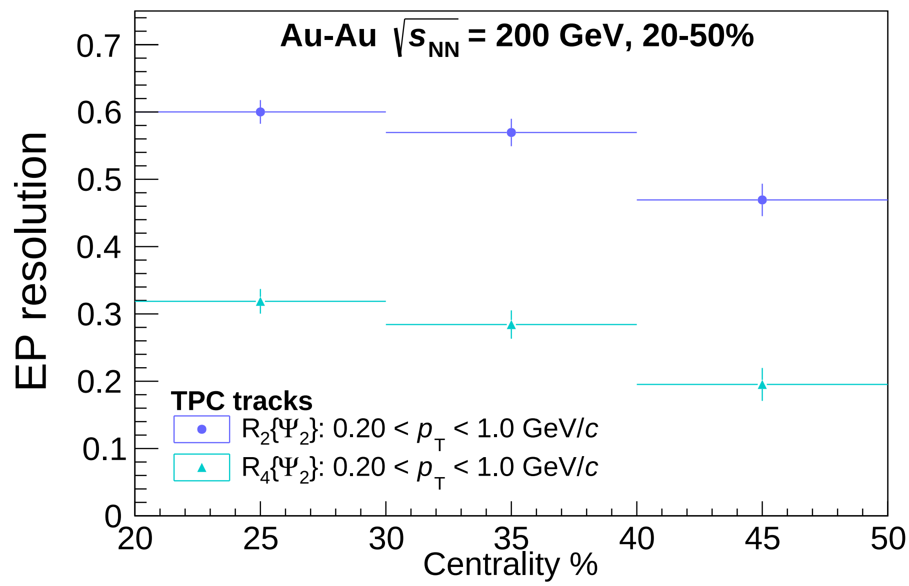


Figure 4.1: Event-plane resolution: Second-order (fourth-order) harmonic relative to the event plane, $R_2(\Psi_2)$ ($R_4(\Psi_2)$), respectively. The approach follows the Modified Reaction Plane (MRP) method [2] utilizing the charged tracks of the TPC for event-plane reconstruction and resolution calculation for tracks ranging from 0.2-1.0 GeV/c.

4.1.1 Jet reconstruction and selection

Full jets are reconstructed by measuring charged-tracks in the TPC and collecting neutral-particle information from the BEMC. The anti-k_{T} algorithm [18] implemented through the FastJet package [19] clusters these particles into jets by reconstructing the jet momenta as the quadratic sum of their constituent momenta using a boost-invariant p_{T}^{2} recombination scheme. Tracks used for reconstructing jets are assumed to be pions while the towers to have arisen from massless particles. The location of a jet, described by the 'jet axis', refers to the azimuthal and pseudorapidity coordinates of the centroid of the jet. Jets can further be described by a resolution parameter, r, which is an input into the anti-k_{T} algorithm. The r parameter determines the radial extent of jet constituents about the jet axis given by $\Delta r = \max(\sqrt{\Delta\phi^2 + \Delta\eta^2})$ where $\Delta\phi$ ($\Delta\eta$) is the azimuthal angle (pseudorapidity) of constituents relative to the jet-axis. All jets measured in this work are clustered with resolution parameter of $r = 0.4$. The area of jets, A_{jet}, is found with FastJet using active ghost particles [20]. Partially reconstructed jets at the edge of the acceptance are rejected by applying a fiducial cut, $|\eta_{\mathrm{jet}}| < 1.0 - r$, to assure all jets fall within the acceptance of the detectors.

Jets produced in heavy-ion collisions sit on top of a large amount of underlying event. The jet signal can be found beneath tens to hundreds of particles resulting from various other processes. To reduce the influence of these background particles, this analysis requires tracks (towers) with $p_{\mathrm{T}}(E_{\mathrm{T}}) > 2.0$ GeV/c for jet reconstruction. At RHIC energies this selection reduces the median background energy density per unit A_{jet}, $\langle\rho\rangle$, down to ≈ 0.6 GeV. This high-constituent selection, referred to as a "hard-core" jet selection reduces fluctuations, fake jets, and background jet particles [21]. To further reduce contributions from the background and to match the trigger condition, the jets are required to contain a constituent tower that fired the HT trigger ($E_{\mathrm{T,tower}} > 5.4$ GeV/c). The remaining jets are studied in classes of jets with $15 < p_{\mathrm{T,jet}} < 20$ GeV/c and $20 < p_{\mathrm{T,jet}} < 40$ GeV/c.

4.1.2 Jet-hadron correlation

The measurement of the correlation function (distribution of charged hadrons relative to reconstructed jets) described in Eqn. 4.1 requires several corrections. The correlation function is measured in pseudorapidity ($\Delta\eta$) and azimuth ($\Delta\phi$) as:

$$\frac{1}{N_{\mathrm{trig}}}\frac{d^2N_{\mathrm{assoc,jet}}}{d\Delta\phi d\Delta\eta} = \frac{1}{N_{\mathrm{trig}}}\frac{1}{\epsilon(p_{\mathrm{T,assoc}},\eta_{\mathrm{assoc}})}\frac{1}{a(p_{\mathrm{T,assoc}},\Delta\phi,\Delta\eta)}\left(\frac{d^2N^{\mathrm{meas}}_{\mathrm{assoc,jet}}}{d\Delta\phi d\Delta\eta} - \frac{d^2N^{\mathrm{bkgd}}_{\mathrm{assoc,jet}}}{d\Delta\phi d\Delta\eta}\right). \tag{4.3}$$

$N_{\mathrm{assoc,jet}}$ gives the number of pairs of trigger jets and the associated hadrons, $N^{\mathrm{meas}}_{\mathrm{assoc,jet}}$ and $N^{\mathrm{bkgd}}_{\mathrm{assoc,jet}}$ are the number of pairs measured and the pair characterized as background respectively. $\epsilon(p_{\mathrm{T,assoc}},\eta_{\mathrm{assoc}})$ is the single-track reconstruction efficiency[1]. The pair acceptance, $a(p_{\mathrm{T,assoc}},\Delta\phi,\Delta\eta)$, is calculated from the raw pairs that we measure from a trigger jet associated with charged hadrons from mixed events.

The correlations are determined in bins of centrality, reconstructed trigger-jet transverse momentum ($p_{\mathrm{T,jet}}$), associated-hadron transverse momentum ($p_{\mathrm{T,assoc}}$), and bins of the trigger jet relative to the event plane (in, mid, out, all combined angles) defined in Sect. 4.1. The corrected correlation functions contain a large combinatorial background ($N^{\mathrm{bkgd}}_{\mathrm{assoc,jet}}$), which must be subtracted. This subtraction procedure is described in Sect. 4.1.3 .

The pair-acceptance correction, $1/a(p_{\mathrm{T,assoc}},\Delta\phi,\Delta\eta)$, accounts for the detector and tracking inefficiencies, and is found by correlating jets from HT-triggered events with associated hadrons from MB events of the same event class. In addition to providing the acceptance correction, the mixed-event procedure will also help remove the trivial correlation due to an η dependence in the single-particle track distributions [4]. Taking the ratio of the single-event pairs by the mixed-event pairs removes the detector effects of acceptance and efficiency. The pair acceptance will serve as the dominant effect, given that there is

[1] Acceptance effects such as gaps between sector boundaries are included in this correction, however, we refer to $\epsilon(p_{\mathrm{T,assoc}},\eta_{\mathrm{assoc}})$ as an efficiency correction in order to distinguish it from $a(p_{\mathrm{T,assoc}},\Delta\phi,\Delta\eta)$.

little η dependence in both the tracks and jets across the acceptance range of this analysis.

The event-mixing procedure used in this analysis is well described in Ref. [17]. The mixed events used for calculating $a(p_{\mathrm{T,assoc}}, \Delta\phi, \Delta\eta)$ in this work are required to be within the same 10% centrality class and to have a vertex position within 4 cm along the direction of the beam (z). They are constructed separately for 20-30%, 30-40%, and 40-50% centrality classes and combined accordingly. High-momentum tracks are nearly straight, so the detector acceptance does not change significantly at high-$p_{\mathrm{T,assoc}}$, and thus all associated momentum bins greater than 2.0 GeV/c are combined to increase statistics. There is no difference in efficiency and acceptance within uncertainties for different orientations of the jet relative to the event plane, and therefore the same correction $a(p_{\mathrm{T,assoc}}, \Delta\phi, \Delta\eta)$ is applied for all angles relative to the event plane. The acceptance $a(p_{\mathrm{T,assoc}}, \Delta\phi, \Delta\eta)$ is normalized to 1 at its maximum, determined using the region of approximately constant acceptance ($|\Delta\eta| < 0.4$). For each $p_{\mathrm{T,assoc}}$ bin, the projection of the flat plateau region (integrated over the region $|\Delta\eta| < 0.4$) was fit with a constant over the whole $\Delta\phi$ range. The associated uncertainties of the fits were used for the systematic uncertainty on the mixed-event normalization, which is added in quadrature and reported as the scale uncertainty. This systematic uncertainty is under 1% (1.25%) in all $p_{\mathrm{T,assoc}}$ bins used for the reported results of jets with $15 < p_{\mathrm{T,jet}} < 20$ GeV/c and $20 < p_{\mathrm{T,jet}} < 40$ GeV/c.

A shape uncertainty in $\Delta\phi$ due to the z_{vtx} binning, similar to that in $\Delta\eta$, could lead to an additional shape uncertainty in the correlation functions. To test for this, the ratio of the 1D $\Delta\phi$ projection with the nominal z_{vtx} binning and the binning described above was calculated for each $p_{\mathrm{T,jet}}$, $p_{\mathrm{T,assoc}}$, and centrality bin. The variations about 1.0 are smaller than the statistical errors associated with the points. This uncertainty was therefore considered negligible.

There is an additional shape uncertainty associated with the application of the acceptance correction due to slight changes in the correlation function at large $\Delta\eta$ in the acceptance with v_z position. The background level is determined from the level of the correlation function at large $\Delta\eta$, leading to a scale uncertainty in the background subtraction dependent on $p_{\mathrm{T,assoc}}$. This uncertainty is from the differences between the nominal (z-integrated) and the z-vertex method (z-binned) for correcting the mixed events on the level of the background in the $0.6 < \Delta\eta < 1.2$ range, and signal plus background, in the $|\Delta\eta| < 0.6$ range. The large $\Delta\eta$ region is used to determine the background so any uncertainties in the level of the correlation function in this region lead to an uncertainty in the level of the background in the signal region. This is expressed as an additional scale band on the final results. This uncertainty is determined by varying the binning of the mixed events in v_z and is correlated for different angles relative to the event plane and for different bins in $p_{\mathrm{T,assoc}}$. Above $p_{\mathrm{T,assoc}} > 3$ GeV/c, this uncertainty is negligible because the background is small.

4.1.3 Flow modulation of combinatorial background

The combinatorial background ($N^{\mathrm{bkgd}}_{\mathrm{assoc,jet}}$) from Eq. 4.3 can be parametrized for trigger jets restricted to the orientation $\mathfrak{R}$ = in/mid/out-of-plane relative to the event plane as [11, 22]:

$$\left(\frac{1}{\pi}\frac{dN^{\mathrm{bkgd}}_{\mathrm{assoc,jet}}}{d\Delta\phi}\right)_{\mathfrak{R}} = B^{\mathfrak{R}}\left(1+\sum_{n=2}^{\infty}2v_{n,\mathrm{assoc}}v^{\mathfrak{R}}_{n,\mathrm{jet}}\cos(n\Delta\phi)\right), \tag{4.4}$$

where $v_{n,\mathrm{assoc}}$ and $v^{\mathfrak{R}}_{n,\mathrm{jet}}$ are the Fourier coefficients of the azimuthal angle distribution (flow coefficients) of background associated particles and trigger jets restricted to the orientation $\mathfrak{R}$ respectively. $B^{\mathfrak{R}}$ gives the background level amplitude for orientation $\mathfrak{R}$. The trigger jets being restricted to orientation $\mathfrak{R}$ modifies their nominal flow coefficients $v_{n,\mathrm{jet}}$'s into the $v^{\mathfrak{R}}_{n,\mathrm{jet}}$ from Eq. 4.4 which are related to each other as [11]:

$$v^{\mathfrak{R}}_{n,\mathrm{jet}} = \begin{cases} \frac{1}{\beta^{\mathfrak{R}}}\left(v_{n,\mathrm{jet}} + \cos\left(n\phi^{\mathfrak{R}}_s\right)\frac{\sin(nc)}{nc}R_n + \sum_{k=2,4,6,..}\left(v_{(k+n),\mathrm{jet}} + v_{|k-n|,\mathrm{jet}}\right)\cos\left(k\phi^{\mathfrak{R}}_s\right)\frac{\sin(kc)}{kc}R_n\right) & \text{if } n \text{ is eve} \\ v_{n,\mathrm{jet}} & \text{if } n \text{ is od} \end{cases} \tag{4.5}$$

Where, $\beta^{\mathfrak{R}}$ is the $\mathfrak{R}$ dependence of $B^{\mathfrak{R}}$ given by,

$$B^{\mathfrak{R}} \propto \beta^{\mathfrak{R}} = 1 + \sum_{k=2,4,6,..} 2v_{k,\mathrm{jet}} \cos\left(k\phi_s^{\mathfrak{R}}\right) \frac{\sin(kc)}{kc} R_n \tag{4.6}$$

$\phi_S^{\mathfrak{R}}$ and c and are the center and width of the $|\Psi_{2,EP} - \phi_{\mathrm{jet}}|$ range for jets restricted to orientation $\mathfrak{R}$. Odd $v_{n,\mathrm{jet}}$'s mainly arise from initial state fluctuations and are therefore uncorrelated with the second-order event-plane and remain constant when the trigger jet is moved relative to the event plane, while even $v_{n,\mathrm{jet}}$'s will change [22, 23]. The $\mathfrak{R}$ dependent background shape is dependent upon the event-plane resolution (R_n), which is fixed at the measured values. Extended details into the derivations of relevant equations can be found in Refs. [11, 24, 25].

Collective particle flow plays a major role in understanding the underlying-event background. This background consists of particles created from mechanisms unrelated to the hard process that led to a jet. Some of the jet signal is correlated with our event plane due to the path-length dependence of partonic energy loss, while soft hadrons are predominantly correlated with the event plane due to hydrodynamical flow that also contribute to the bulk-particle production.

To remove the combinatorial background comprised by contributions from the underlying event, the reaction plane fit (RPF) developed in Ref. [22] is applied in this work. The measured jet-hadron correlation signal is decomposed into a near-side and an away-side, with the former being narrow in $\Delta\phi$ and $\Delta\eta$ and the latter narrow in $\Delta\phi$, but broad in $\Delta\eta$. The narrowness of the near-side implies the signal is negligible at large $\Delta\eta$, where the background dominates. Applying RPF, we defined our 'signal + background' region for $|\Delta\eta| < 0.6$ and further $0.6 \leq |\Delta\eta| < 1.2$ as a background dominated region where the signal was assumed to be negligible. The correlation function is fit with the parametrization given in Eq. 4.4, restricted to $n = 4$ for the background dominated region at large $\Delta\eta$ and small $\Delta\phi$ ($|\Delta\phi| < \pi/2$) simultaneously for in-plane , mid-plane , and out-of-plane trigger jets.

RPF improves upon prior background subtraction techniques [4, 26] by avoiding problems due to contamination from jets on both the near- and away-side by using the near-side at large $\Delta\eta$ and also using the dependence of the flow-modulated background on the angle of the trigger jet relative to the event plane to constrain the background shape and level.

For $p_{\mathrm{T,assoc}} > 2$ GeV/c, the combinatorial background is small, and few high momentum tracks are found at large distances in pseudorapidity from the near-side jet. So the model fit is restricted to $n = 3$ as higher order terms are no longer contributing. This occurs for $p_{\mathrm{T,assoc}} > 4$ GeV/c with $15 < p_{\mathrm{T,jet}} < 20$ GeV/c jets and $p_{\mathrm{T,assoc}} > 3$ GeV/c with $20 < p_{\mathrm{T,jet}} < 40$ GeV/c. Therefore, the RPF fits consist of six parameters ($B^{\mathfrak{R}}$, $v_{2,\mathrm{jet}}^{\mathfrak{R}}$, $v_{2,\mathrm{assoc}}$, ($v_{3,\mathrm{jet}} \times v_{3,\mathrm{assoc}}$), $v_{4,\mathrm{jet}}^{\mathfrak{R}}$, and $v_{4,\mathrm{assoc}}$) for low $p_{\mathrm{T,assoc}}$, and four ($B^{\mathfrak{R}}$, $v_{2,\mathrm{jet}}^{\mathfrak{R}}$, $v_{2,\mathrm{assoc}}$ and ($v_{3,\mathrm{jet}} \times v_{3,\mathrm{assoc}}$)) for high $p_{\mathrm{T,assoc}}$.

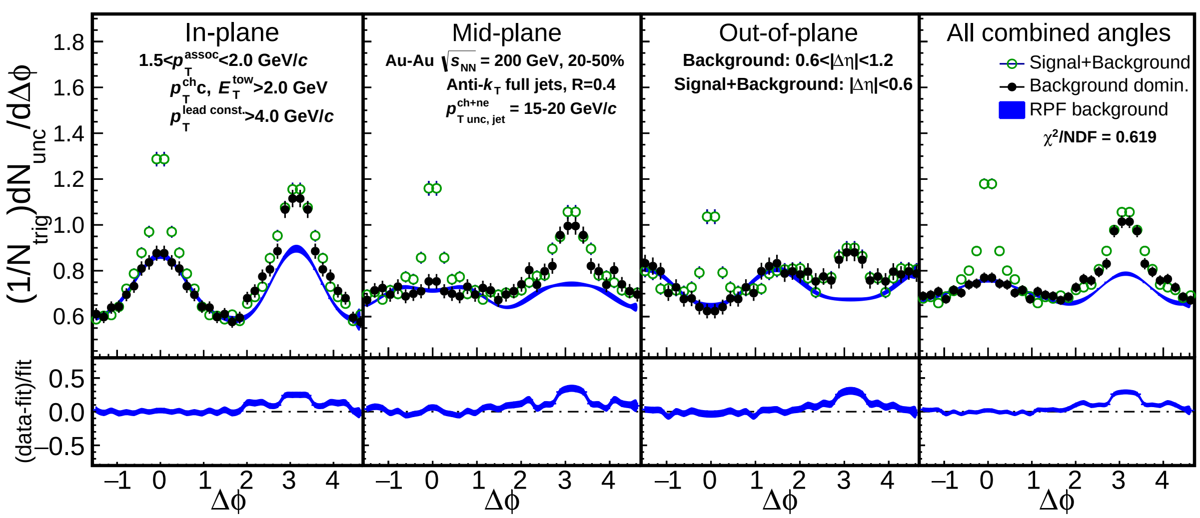


Figure 4.2: (Top) Signal+background region, background-dominated region, and RPF fit to the background for the event-plane dependent jet-hadron correlations of $15 < p_{\mathrm{T}}^{\mathrm{jet}} < 20$ GeV/c jets correlated with $1.5 < p_{\mathrm{T,assoc}} < 2.0$ GeV/c charged hadrons from the 20-50% most central events. (Bottom) Quality of the RPF fit to the background-dominated region expressed as (data - fit) / fit.

Comparison of in-, mid-, and out-of-plane jet-hadron correlations is performed after background subtraction to explore the effects related to event-plane orientation. An example of event-plane dependent correlation function after the RPF background subtraction for jets

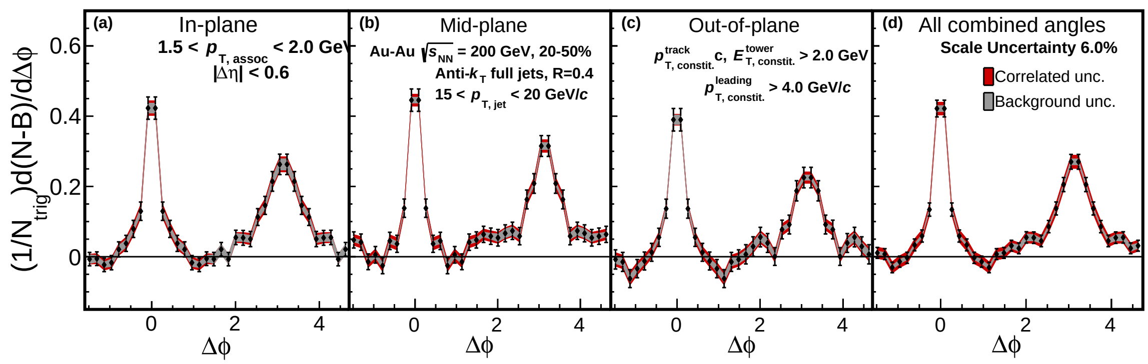


Figure 4.3: Example of a background subtracted correlation function for $15 < p_{\mathrm{T,jet}} < 20$ GeV/c jets correlated with $1.5 < p_{\mathrm{T,assoc}} < 2.0$ GeV/c charged hadrons from the 20-50% most central collisions. Correlated scaled uncertainty from the application of mixed events is displayed by the red band, while the uncertainty associated with the RPF background fit is displayed as the gray band.

with $15 < p_{\mathrm{T,jet}} < 20$ GeV/c is shown in Fig. 4.3 for in-plane (a), mid-plane (b), out-of-plane (c) and jets from all combined angles (d) for associated particles with momenta $1.5 < p_{\mathrm{T,assoc}} < 2.0$ GeV/c. The uncertainties from the RPF background subtraction are propagated using the covariance matrix from the fit and are non-trivially correlated point-to-point and between different bins relative to the event plane. These are shown in gray. The uncertainty from the acceptance correction, described in Sect. 4.1.2 is displayed by the red uncertainty band. The uncertainties on the event-plane resolution are negligible relative to that of the background subtraction and statistical uncertainties of the final results. Additional uncertainties uncorrelated with each other, but correlated for all points in $\Delta\phi$ are given in Tab. 4.1. They are combined in quadrature and listed as the scale uncertainty on the results.

4.1.4 Systematic Uncertainties

The PYTHIA6 Perugia 2012 tune is used to create particle level dijet events embedded in MB Au+Au 200 GeV events at the detector level. This allows for further comparisons between the jets from the input PYTHIA6 tracks (generator-level jets or GEN-jets) and the jets from the embedded tracks reconstructed by GEANT (reconstructed jets or RECO-jets).

RECO level events are analyzed with the same selections and parameters as used by the data analysis. All particles of GEN-level events are required to be in their final state (particles with no further daughters). GEN-level jets have $p_{\mathrm{T,jet}}^{\mathrm{GEN}} > 10$ GeV/c and the RECO-level jets have $p_{\mathrm{T,jet}}^{\mathrm{RECO}} > 5$ GeV/c. Further, we require only the RECO-level jets contain a neutral component with $E_{\mathrm{T}} \geq 5.4$ GeV to match the trigger condition applied in data. Our goal is to study the effects of the detector response on jet reconstruction and our analysis. In order to compare the same jet pre- and post-reconstruction, we apply a 'nearness' criteria in the $\eta - \phi$ that matches a GEN-jet to one RECO-jet satisfying:

(a) $r_{\mathrm{GEN,RECO}} = \sqrt{(\eta_{\mathrm{jet}}^{\mathrm{GEN}} - \eta_{\mathrm{jet}}^{\mathrm{RECO}})^2 + (\phi_{\mathrm{jet}}^{\mathrm{GEN}} - \phi_{\mathrm{jet}}^{\mathrm{RECO}})^2} \leq 0.4$, $r_{\mathrm{GEN,RECO}}$ being the separation between the gen-jet and the reco-jet in $\eta - \phi$ space, (b) the RECO-jet is the closest to the gen-jet among all reco-jets (minimize $r_{\mathrm{GEN,RECO}}$).

We compare the resulting GEN- and RECO-level jet spectra to calculate the momentum resolution which is given by:

$$p_{\mathrm{T,jet}} \text{ resolution}(\%) = \frac{p_{\mathrm{T,jet}}^{\mathrm{GEN}} - p_{\mathrm{T,jet}}^{\mathrm{RECO}}}{p_{\mathrm{T,jet}}^{\mathrm{GEN}}} \times 100\%. \tag{4.7}$$

Fig. 4.4 shows the $p_{\mathrm{T,jet}}$ resolution for $15 < p_{\mathrm{T,jet}}^{\mathrm{GEN}} < 20$ GeV/c (left) and $20 < p_{\mathrm{T,jet}}^{\mathrm{GEN}} < 40$ GeV/c (right) $R = 0.4$ full jets. The distributions have been normalized into probability functions and are shown for the 20-50% most central events for all angles of the jet relative to the event plane. Fig. 4.4 further shows an average energy loss of around 15% going from GEN to RECO. This net energy shift is thought to be due to counteracting effects of the tracking inefficiency at the RECO level and there being more tracks in the RECO level from the min-bias pedestal. The $p_{\mathrm{T,jet}}$ resolution for $15 < p_{\mathrm{T,jet}}^{\mathrm{RECO}} < 40$ GeV/c reco-jets is roughly 10-20%. The event-plane dependence of the $p_{\mathrm{T,jet}}$ resolution was also studied and found to be within 1-2% of each other between different orientations of jets w.r.t event plane. There can be slight differences in the jets reconstructed at lower momenta with $15 < p_{\mathrm{T,jet}} < 20$ GeV/c for jets at different angles relative to the event plane, due to a low momentum embedded jet overlapping with another jet in the Au+Au data and from statistical fluctuations. As there are more jets from in-plane than out-of-plane orientations in the data, this leads to an apparent difference in the reconstructed jet spectra. Otherwise there are no significant differences between jets at different angles relative to the event plane. A map between $p_{\mathrm{T,jet}}^{\mathrm{GEN}}$ and $p_{\mathrm{T,jet}}^{\mathrm{RECO}}$, called the response matrix, is added to Fig. 4.4 as a smaller inset on the top right.

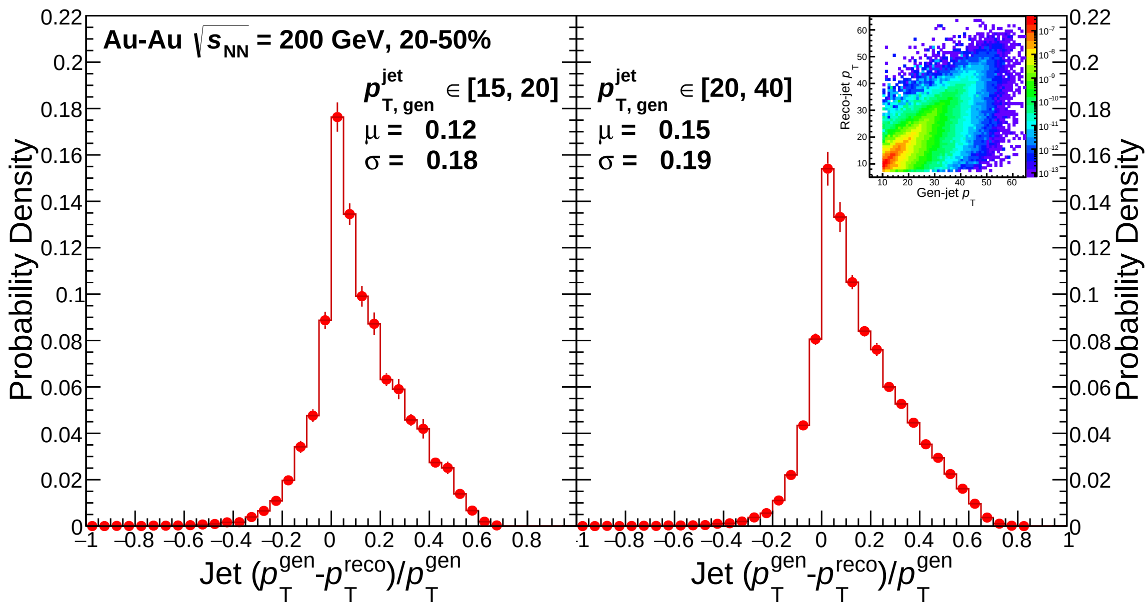


Figure 4.4: $p_{\mathrm{T,jet}}$ resolution for $15 < p_{\mathrm{T,jet}}^{\mathrm{GEN}} < 20$ GeV/c (left) and $20 < p_{\mathrm{T,jet}}^{\mathrm{GEN}} < 40$ GeV/c (right) $R = 0.4$ full jets with the corresponding response matrix as an inset on the top right. Jets are measured from all angles relative to the event plane in the 20-50% most central events. This comparison is for matched GEN-level to RECO-level jets.

We summarize the systematic uncertainties in Tabs. 4.1, 4.2, and 4.3. Table 4.1 lists the sources of systematic uncertainties which are independent of the angle relative to the event plane. These sources include the single-track reconstruction efficiency (Sect. **??**), uncertainties in the mixed events due to their normalization and shape in $\Delta\phi$ (Sect. 4.1.2), and uncertainties in the event-plane resolution (Sect. 2.2.1). The uncertainties are added in quadrature and lead to a 6% uncertainty in the scale of the correlation functions and yields with the single-track reconstruction efficiency being the dominate source. This uncertainty is uncorrelated for different associated-particle momenta.

Table 4.1: Summary of systematic uncertainties which are independent of the angle relative to the event plane and the momentum for both $15 < p_{\mathrm{T,jet}} < 20$ GeV/c and $20 < p_{\mathrm{T,jet}} < 40$ GeV/c in 20-50% central Au+Au collisions.

Source	Uncertainty %
Single particle reconstruction efficiency	5
Mixed event (shape $\Delta\phi$)	negligible
Mixed event normalization	< 1.25
Event-plane resolution	< 1

Additional uncertainties, highly dependent on the angle of the jet relative to the event plane and the associated particles' momentum, are summarized in Tabs. 4.2 and 4.3. These include the impact of the scale uncertainty from the mixed events (Sect. 4.1.2) and the RPF background fit (Sect. 4.1.3) on the associated yield and jet-peak width results. These uncertainties are compared for two associated particle momentum bins (1.0-1.5 and 3.0-4.0 GeV/c) to highlight how much of an impact the background has at low momenta. The uncertainty arising from the RPF background subtraction is non-trivially correlated point-to-point in $\Delta\phi$ and for different orientations of the jet relative to the event plane, but is uncorrelated between different $p_{\mathrm{T,assoc}}$ bins. The acceptance-correction uncertainty on the shape in $\Delta\eta$ is also correlated for different orientations of the jet relative to the event plane and uncorrelated across $p_{\mathrm{T,assoc}}$ bins. The acceptance-shape uncertainty is the dominant source at low-momenta, while being more comparable to the background uncertainty at larger momenta.

To account for jet-energy shift (JES) due to underlying-event background contribution to the reconstructed jet energy and to provide an adequate comparison for the jet samples from different orientations relative to event plane, a random-cone study was performed on the minimum bias data of matching centrality selection. The $\langle\rho\rangle$ is found to be 0.388 GeV/c (in-plane), 0.344 GeV/c (mid-plane), and 0.308 GeV/c (out-of-plane), with a corresponding RMS of 0.4 GeV/c. Shifting the jet-momenta selection by the corresponding thresholds was utilized in the correlation analysis to remove the shift-related differences between the jet collections.

Table 4.2: Summary of systematic uncertainties on the associated yields and widths calculated from the correlation functions due to the shape uncertainty coming from the shape of the acceptance correction in $\Delta\eta$, the correlated background-fit uncertainty, and the uncertainty associated with the jet energy shift (JES) correction, each varying with event-plane orientation bins. They are displayed for $15 < p_{\mathrm{T,jet}} < 20$ GeV/c in 20-50% central Au+Au collisions for $1.0 < p_{\mathrm{T,assoc}} < 1.5$ GeV/c and $3.0 < p_{\mathrm{T,assoc}} < 4.0$ GeV/c bins. The values are expressed as a percent of the nominal value.

Source	Result	Orientation	Uncertainty %			
			Near-side: $p_{\mathrm{T,assoc}}$ (GeV/c)		Away-side: $p_{\mathrm{T,assoc}}$ (GeV/c)	
			1.0-1.5	3.0-4.0	1.0-1.5	3.0-4.0
Acceptance shape	Yield	in-plane	14	1.2	8.1	3.0
		mid-plane	11	1.2	7.6	3.3
		out-of-plane	11	1.1	7.3	3.0
	Width	in-plane	5.2	0.6	4.3	2.4
		mid-plane	5.4	0.5	3.9	2.2
		out-of-plane	4.3	0.4	4.9	2.1
Background fit	Yield	in-plane	11	0.8	6.1	2.1
		mid-plane	7.7	0.9	5.2	2.5
		out-of-plane	7.4	0.7	4.7	2.1
	Width	in-plane	10	0.1	8.2	0.4
		mid-plane	10	0.1	7.5	0.4
		out-of-plane	8.2	0.1	9.3	0.4
JES correction	Yield	in-plane	1.8	3.7	4.9	4.0
		mid-plane	2.8	2.5	1.4	4.2
		out-of-plane	3.7	3.3	4.8	4.2
	Width	in-plane	0.6	0.4	4.0	< 0.1
		mid-plane	0.2	< 0.1	7.9	1.5
		out-of-plane	1.6	0.7	1.9	1.7

Table 4.3: Summary of systematic uncertainties on the associated yields and widths calculated from the correlation functions due to the shape uncertainty coming from the shape of the acceptance correction in $\Delta\eta$, the correlated background-fit uncertainty, and the uncertainty associated with the JES correction, each varying with event-plane orientation bins. They are displayed for $20 < p_{\mathrm{T,jet}} < 40$ GeV/c in 20-50% central Au+Au collisions for $1.0 < p_{\mathrm{T,assoc}} < 1.5$ GeV/c and $3.0 < p_{\mathrm{T,assoc}} < 4.0$ GeV/c bins. The values are expressed as a percent of the nominal value.

Source	Result	Orientation	Uncertainty %			
			Near-side: $p_{\mathrm{T,assoc}}$ (GeV/c)		Away-side: $p_{\mathrm{T,assoc}}$ (GeV/c)	
			1.0-1.5	3.0-4.0	1.0-1.5	3.0-4.0
Acceptance shape	Yield	in-plane	35	1.2	22	2.5
		mid-plane	39	1.2	22	2.2
		out-of-plane	59	0.9	37	1.6
	Width	in-plane	30	0.7	8.4	1.9
		mid-plane	22	0.5	15	1.7
		out-of-plane	19	0.4	28	1.2
Background fit	Yield	in-plane	13	1.0	8.2	2.1
		mid-plane	14	0.7	8.0	1.2
		out-of-plane	22	0.9	14	1.5
	Width	in-plane	13	0.1	3.7	0.2
		mid-plane	10	0.1	6.8	0.2
		out-of-plane	8.6	< 0.1	13	0.1
JES correction	Yield	in-plane	0.2	0.4	< 0.1	2.6
		mid-plane	1.1	0.3	2.1	2.9
		out-of-plane	3.1	0.6	1.4	3.5
	Width	in-plane	1.2	0.5	3.3	0.3
		mid-plane	1.2	0.7	1.3	1.5
		out-of-plane	2.2	0.1	0.2	1.9

4.2 Results

Charged particle yields associated with jets are found by integrating the associated yield:

$$\text{Yield} = \frac{1}{N_{\text{trig}}} \int_{c}^{d} \int_{a}^{b} \frac{d^2 N_{\text{assoc,jet}}}{d\Delta\phi d\Delta\eta} d\Delta\phi d\Delta\eta. \tag{4.8}$$

The integration limits in $\Delta\phi$ for the near-side are $a = -\pi/3$ and $b = \pi/3$, while for the away-side, we have $a = 2\pi/3$ and $b = 4\pi/3$. The integration limits in $\Delta\eta$ are the same for both the near-side and away-side, $c = -0.6$ and $d = 0.6$.

Shown in Fig. 4.5 is the near-side (left) and away-side (right) associated yields vs $p_{\text{T,assoc}}$ for $15 < p_{\text{T,jet}} < 20$ GeV/c (top) and $20 < p_{\text{T,jet}} < 40$ GeV/c GeV/c (bottom) full jets in 20-50% centrality collisions. The yields are compared for each orientation of the trigger jet reconstructed relative to the event plane (in/mid/out) for $p_{\text{T,assoc}} \in$ [1.0, 1.5], [1.5, 2.0], [2.0, 3.0], [3.0, 4.0], [4.0, 6.0], [6.0, 10.0] GeV/c.

The main feature is the steeply falling associated yield with increasing $p_{\text{T,assoc}}$, which occurs on both the near- and away-side. Note that associated yields for $p_{\text{T,assoc}} \geq 2.0$ GeV/c include jet constituents, which leads to the discontinuity at $p_{\text{T,assoc}} = 2$ GeV/c. Additionally, the use of jets with 'hard-cores' and containment of a tower associated to the firing event trigger can lead to a surface bias of the near-side jet. This, however, maximizes the average path length the away-side recoil jets travels, increasing the likelihood of an interaction with the medium. We would expect an in-plane jet and an out-of-plane jet with the same $p_{\text{T,jet}}$ to have different distributions of hard and soft constituents. An in-plane jet with less path travelled in the medium on average would be expected to show higher yields of constituents with higher $p_{\text{T,assoc}}$ than the more quenched out-of-plane jet, which would be expected to show higher yields of constituents with lower $p_{\text{T,assoc}}$. While uncertainties are smaller for jets with $15 < p_{\text{T,jet}} < 20$ GeV/c, both samples still lack any clear dependence on the event-plane angle. This is an indication that modifications dependent on the average path length are smaller than the experimental uncertainties. On the far right of the near- and away-side panels is the inclusive $p_{\mathrm{T,assoc}}$ bin with $1.0 < p_{\mathrm{T,assoc}} < 10$ GeV/c. The associated yields of each event-plane orientation in the inclusive $p_{\mathrm{T,assoc}}$ selection are consistent with each other for the sample where $15 < p_{\mathrm{T,jet}} < 20$ GeV/c. However, in the jet sample with $20 < p_{\mathrm{T,jet}} < 40$ GeV/c, there are indications suggesting potential modifications. This potential modification is apparent on both the near- and away-side with the largest contributions coming from the lowest $p_{\mathrm{T,assoc}}$ bins.

The widths are calculated by fitting a Gaussian, $Ae^{(\Delta\phi-\Delta\phi_0)^2/2\sigma^2}$, to the jet peak centered at $\Delta\phi_0 = 0$ for the near-side and $\Delta\phi_0 = \pi$ for the away-side. The gaussians are fitted separately, with a range of $|\Delta\phi| < \pi/3$ on the near-side and $|\Delta\phi - \pi| < \pi/3$ on the away-side. The Gaussian fit is repeated with different values of the background parameters and the covariance matrix is used to propagate the uncertainties. The scale uncertainties on the widths are given by $\sigma_w^{sc} = \frac{B}{\sigma_B} \times |\alpha - 1|$, where α is the p_{T}-dependent scale factor associated with the acceptance shape when propagating the background determined in the $0.6 \leq |\Delta\eta| < 1.2$ to the region $|\Delta\eta| < 0.6$.

From Fig. 4.6, it is clear a broadening of the jet peaks is occurring for decreasing $p_{\mathrm{T,assoc}}$. This is expected from both collisional energy loss and gluon bremsstrahlung. With out-of-plane jets expected to traverse a longer average path length than in-plane jets, this would lead to additional interactions with the medium and more subsequent re-scatterings resulting in a larger width for jets out-of-plane relative to in-plane. Within the uncertainties there is no clear ordering. This indicates the effect of path-length dependent energy loss is not large enough to be seen by the current precision of the data. On the far right of the near- and away-side panels is the inclusive $p_{\mathrm{T,assoc}}$ bin with $1.0 < p_{\mathrm{T,assoc}} < 10$ GeV/c. The widths of each event-plane orientation in the inclusive selection for the the sample with $15 < p_{\mathrm{T,jet}} < 20$ GeV/c are consistent. Conversely, the $15 < p_{\mathrm{T,jet}} < 20$ GeV/c sample reveals indications of potential modifications. This potential modification is apparent on both the near- and away-side and primarily comes from the lowest transverse momentum bins where sample size is the largest.

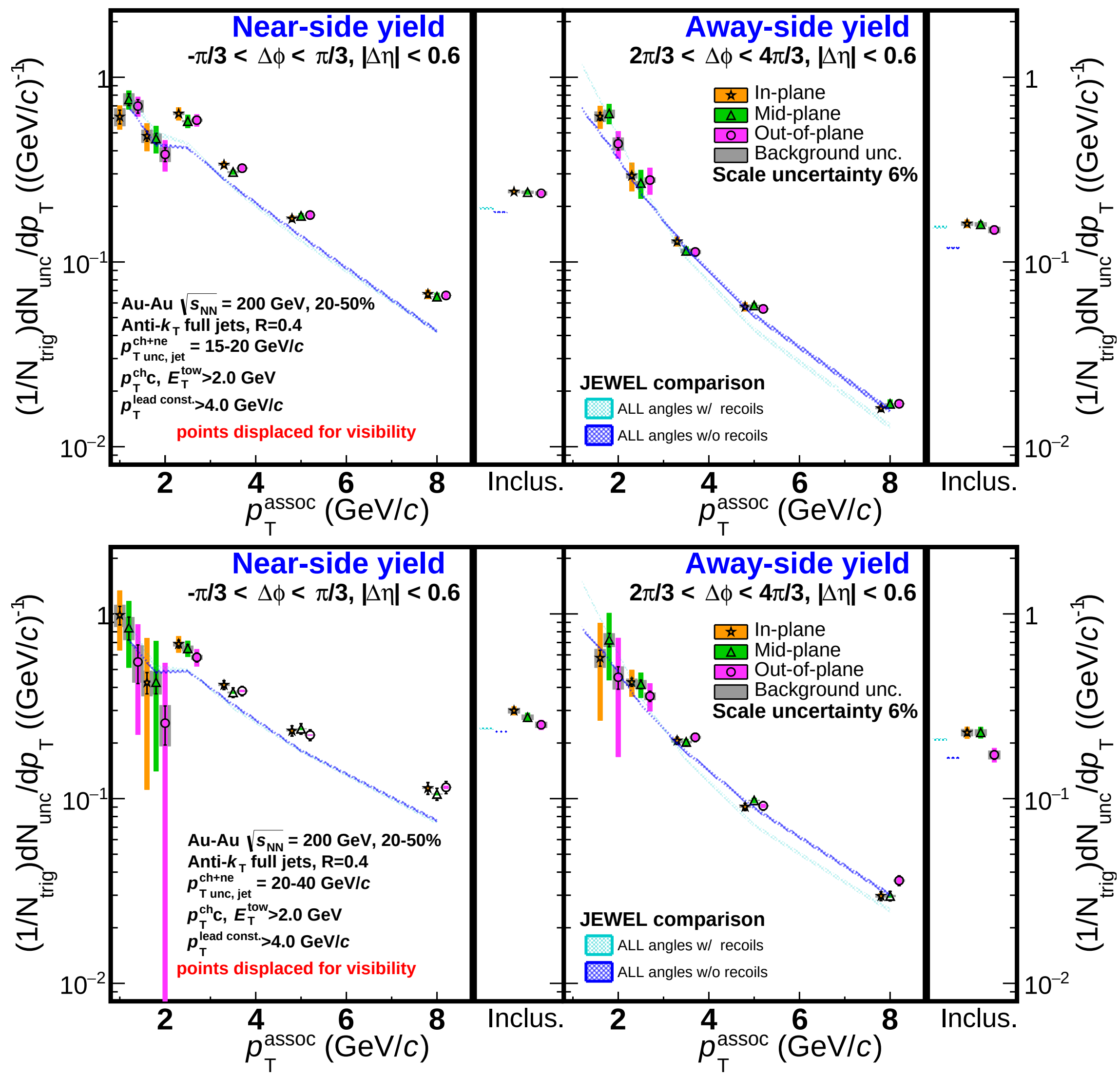


Figure 4.5: Near-side (left) and away-side (right) uncorrected (unc.) associated yield vs $p_{\mathrm{T,assoc}}$ for 15-20 (top) and 20-40 GeV/c (bottom) full jets of 20-50% centrality in Au+Au collisions. The grey bands describe the systematic uncertainties of the background fits and are non-trivially correlated point-to-point. The colored bands are scale uncertainties from the mixed event acceptance shape and JES correction. There is an additional 6% global scale uncertainty. Included on the far right of the near- and away-side panels is the inclusive transverse momentum bin from 1.0-10.0 GeV/c. Points are displaced for visibility.

The measurements presented in Figs. 4.5 and 4.6 are compared to calculations from "Jet Evolution With Energy Loss" known as JEWEL [27], a jet energy loss model based on radiative and collisional energy loss in connection with partons sampled from a longitudinally expanding medium [28]. Model calculations are provided for two regimes, for calculations

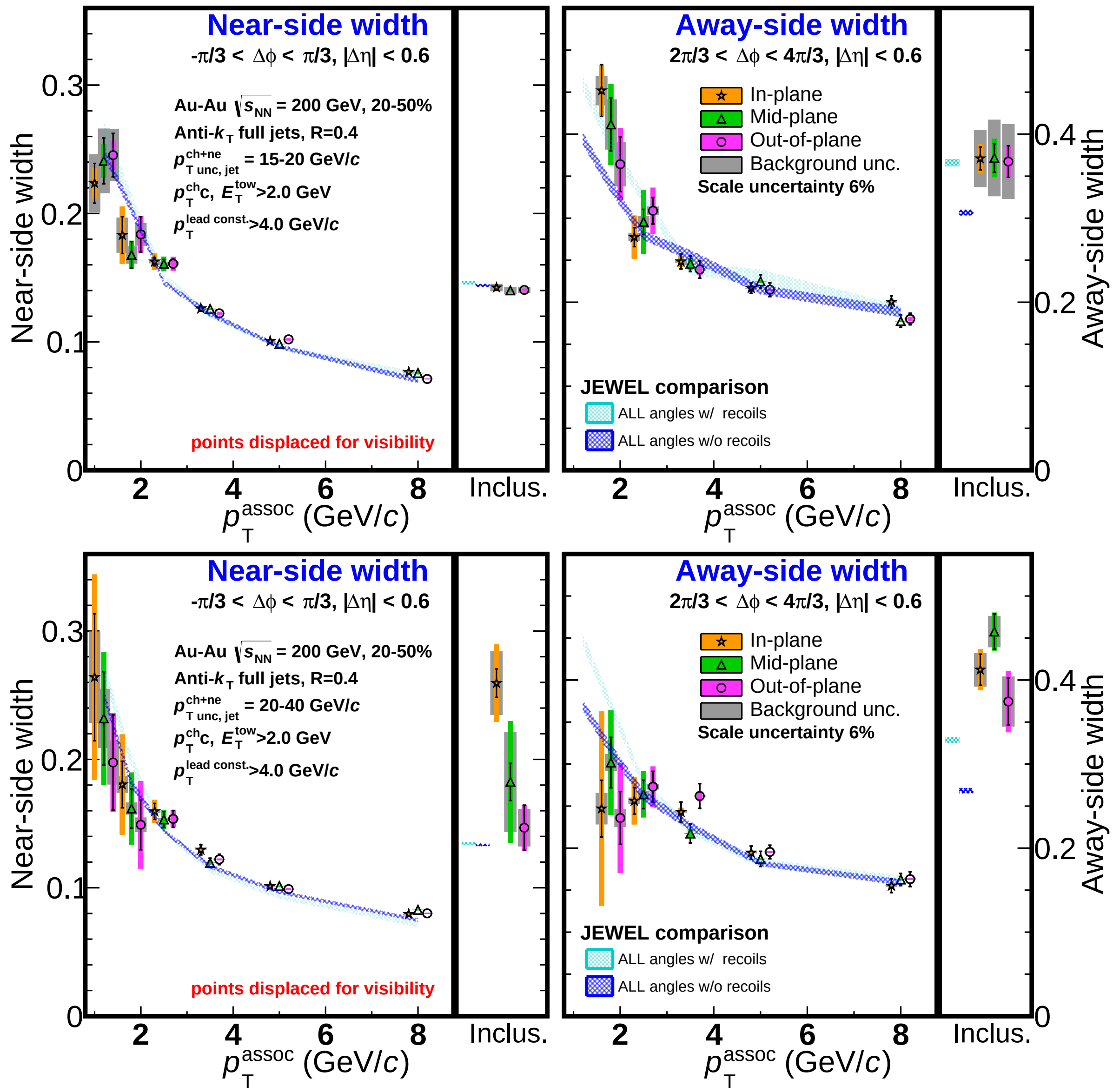


Figure 4.6: Near-side (left) and away-side (right) jet width vs $p_{\mathrm{T,assoc}}$ for 15-20 (top) and 20-40 GeV/c (bottom) full jets of 20-50% centrality in Au+Au collisions. The widths are extracted from the gaussian fit to the jet peak. The grey bands describe the systematic uncertainties of the background fits which are non-trivially correlated point-to-point. The colored bands are scale uncertainties from the mixed-event acceptance shape and JES correction. There is an additional 6% global scale uncertainty. Included on the far right of the near- and away-side panels is an inclusive transverse momentum bin from 1.0-10.0 GeV/c. Points are displaced for visibility.

that a) include recoiled partons, and b) do not include recoiled partons. When no recoils are kept, the lost jet momentum is removed from the entire system. This is useful for modeling energy loss in the hard part of the jet. When recoils are kept, the jets momentum is fully conserved, but this adds both energy and additional background particles to the di-jet. In an experimental analysis, we likely would measure some, but not all of the recoil particles as they are often indistinguishable from background.

Due to the dominant impact of jet-by-jet fluctuations on partonic energy loss over path-length dependence [29, 30], JEWEL only predicts a very slight event-plane dependence which is well below the systematic uncertainty in the measurement. Variations among event-plane orientations were not seen at the 10% level. This is therefore consistent with path-length dependence having an insignificant impact compared to jet-by-jet fluctuations in energy loss. Fluctuations in the density of the medium may also suppress observable path-length dependence and are not included in the JEWEL model. However, higher precision JEWEL calculations may be needed to discern any potential event-plane dependent effects. We thus show the JEWEL comparisons corresponding to the sample integrated over all angles relative to the event plane and compare to the results from data. Comparisons show that the away-side is well described in terms of both the associated yields and widths by including recoils at low $p_{\mathrm{T,assoc}}$, while at high $p_{\mathrm{T,assoc}}$, the yields are better described by not including recoils and the widths have similar results to within uncertainties for both cases. When looking at the near-side, the widths are quite similar at high $p_{\mathrm{T,assoc}}$ with slightly larger values when including recoils at low $p_{\mathrm{T,assoc}}$. For the associated yields at high $p_{\mathrm{T,assoc}}$, both JEWEL cases are similar to each other, but underestimate the data and including recoils has larger yields that better match the data at low $p_{\mathrm{T,assoc}}$.

Alternatively, the initial geometry within specific centrality bins may be fluctuating at a larger magnitude than any possible event-plane dependence [31, 32]. The possibility of this occurrence can be studied by looking at the initial configuration and selecting low and high ellipticity events by using the q_n flow vector found within a selected centrality range.

To study the impact of surface bias and event-plane resolution, a check was performed to investigate the systematic change in the ratio of yields (out/in, mid/in) with the angle of the event plane by fitting a constant to Fig. **??** for jets with $15 < p_{\mathrm{T,jet}} < 20$ GeV/c (top) and $20 < p_{\mathrm{T,jet}} < 40$ GeV/c (bottom). The systematic uncertainties are treated as uncorrelated point-to-point and added to the statistical uncertainties in quadrature. The results are shown in Tabs. 4.4 and 4.5 and are consistent with one. Care should be taken when interpreting the results, as various effects can be in play and medium modifications could give way to a $p_{\mathrm{T,assoc}}$ dependence. This $p_{\mathrm{T,assoc}}$ dependence can be seen as an enhancement of associated yields for $p_{\mathrm{T,assoc}} \geq 2$ GeV/c on the near-side from the geometric kinematic biases due to the “hard-core" requirement of jet reconstruction.

Table 4.4: Results of fits to Fig.**??** (top panel: 15-20 GeV/c jets) to a constant c, the χ^2 over the number of degrees of freedom (NDF), the number of standard deviations σ of c from one, and the range of c within a 90% confidence limit (CL).

	Near-side		Away-side	
parameter	$Y_{\mathrm{out}}/Y_{\mathrm{in}}$	$Y_{\mathrm{mid}}/Y_{\mathrm{in}}$	$Y_{\mathrm{out}}/Y_{\mathrm{in}}$	$Y_{\mathrm{mid}}/Y_{\mathrm{in}}$
c	0.93 ± 0.042	0.949 ± 0.038	0.89 ± 0.065	0.997 ± 0.066
χ^2/NDF	0.68	0.67	0.71	0.18
σ	-1.6	-1.3	-1.7	-0.1
90% CL	0.86 – 1.00	0.89 – 1.01	0.78 – 1.00	0.94 – 1.05

Table 4.5: Results of fits to Fig.**??** (bottom panel: 20-40 GeV/c jets) to a constant c, the χ^2 over the number of degrees of freedom (NDF), the number of standard deviations σ of c from one, and the range of c within a 90% confidence limit (CL).

	Near-side		Away-side	
parameter	$Y_{\mathrm{out}}/Y_{\mathrm{in}}$	$Y_{\mathrm{mid}}/Y_{\mathrm{in}}$	$Y_{\mathrm{out}}/Y_{\mathrm{in}}$	$Y_{\mathrm{mid}}/Y_{\mathrm{in}}$
c	0.874 ± 0.043	0.937 ± 0.043	0.752 ± 0.064	1.02 ± 0.075
χ^2/NDF	1.4	0.19	2.1	0.49
σ	-2.9	-1.5	-3.9	0.2
90% CL	0.77 – 0.98	0.90 – 0.97	0.56 – 0.94	0.91 – 1.12

4.3 Conclusions

The measurement of jet-hadron correlations relative to the event plane is reported for the 20-50% most central events in Au+Au collisions at $\sqrt{s_{\mathrm{NN}}} ==$ 200 GeV in STAR. Partonic interactions are directly related to the distance traversed in the medium, so it is expected that medium-induced jet modifications should depend on the path length. The angle of the jet, measured with respect to the event plane, is correlated on average with the jet's path length through the medium. In this analysis, the average path length of away-side jets is potentially increased due to the surface bias of the near-side trigger jet. This work utilizes the RPF background-subtraction method to remove the event-plane dependent background while reducing uncertainties and assumptions associated with previous background-subtraction techniques. Associated yields, their ratios, and jet-peak widths are extracted for each event-plane orientation and compared with different average path lengths and JEWEL model calculations. JEWEL performs better in describing the associated yields and widths at higher $p_{\mathrm{T,assoc}}$ when recoil partons are not included. Conversely, including recoil partons leads to JEWEL providing a better description of the lower $p_{\mathrm{T,assoc}}$ region. This study highlights the importance of conducting further tuning of Monte Carlo simulations to accurately describe the results in this analysis for jets that are biased towards hard-fragmented jets due to the HT-trigger and hard-core constituent requirements.

Chapter 5

Jet angularities

Within jets originating from the initial stages of hadron collisions, angular scales correlate with temporal and energy scales, giving insight into jet evolution across the perturbative and non-perturbative quantum chromodynamic (p/npQCD) regimes. SoftDrop-grooming [**?**] isolates the pQCD contributions to jet substructure by removing soft, wide-angle radiation. In presence of quark-gluon plasma (QGP), comparing groomed and ungroomed angular distributions to a $p+p$ baseline reveals signatures of jet quenching, such as broadening. They are calculated as:

$$\lambda_\beta^\kappa = \sum_{\text{const}\in\text{jet}} \left(\frac{p_{\text{T,const}}}{p_{\text{T,jet}}}\right)^\kappa r(\text{const}, \text{jet})^\beta, \tag{5.1}$$

where $p_{\text{T,jet}}$ is the jet's total momentum, and $r(\text{const}, \text{jet}) = \sqrt{(\eta_{\text{jet}} - \eta_{\text{const}})^2 + (\phi_{\text{jet}} - \phi_{\text{const}})^2}$ is the (η, ϕ) distance of a constituent from the jet-axis. Parameters κ and β tune experimental sensitivity to hard and wide-angle radiation, respectively. λ_β^1's are infra-red and collinear (IRC) safe angularites [33], which probe the average angular spread of energy around the jet-axis. They are radial moments of the jet's momentum profile, also known as differential jet-shape ($\rho(\mathbf{r})$), given by,

$$\rho(\mathbf{r}) = \lim_{\delta r \to 0} \left\langle \frac{1}{\delta r} \frac{\sum_{|\mathbf{r}_{\text{const}} - \mathbf{r}| < \delta r/2} p_{\text{T,const}}}{p_{\text{T,jet}}} \right\rangle_{\text{jets}}, \tag{5.2}$$

where $\mathbf{r}_{\text{const}} = (\eta_{\text{const}} - \eta_{\text{jet}})\hat{\eta} + (\phi_{\text{const}} - \phi_{\text{jet}})\hat{\phi}$, and it follows that,

$$\lambda_{\beta}^{1} = \int_{\text{jet}} r^{\beta} \rho(\mathbf{r}) d\mathbf{r}. \tag{5.3}$$

The jet angularity based observables like jet-substructure measurements in Pb+Pb collisions at $\sqrt{s_{\text{NN}}} = 2.76$ TeV at the LHC, have shown quenched jets, on average, have migration of charged energy away from their axis relative to a $p+p$ baseline [34] and possibly a survivor bias toward harder, quark-like fragmentation [35]. Similar measurements using jets with lower $p_{\text{T,jet}}$ at RHIC, will help understand jet-medium interactions in a complementary phase-space region to LHC.'

5.1 Dataset and Analysis Method

The analysis uses data from $p+p$ collisions at $\sqrt{s} = 200$ GeV collected in 2012 using the Solenoidal Tracker At RHIC (STAR) detector system. Charged-particle tracks and neutral energy depositions (towers) are measured using STAR's Time Projection Chamber (TPC) [36] and Barrel Electromagnetic Calorimeter (BEMC) [37] detectors respectively. Together, they provide full azimuthal coverage with a pseudorapidity acceptance of $|\eta| \leq 1$. The tracks and towers are clustered into jets using the anti-k_{T} algorithm with a jet resolution parameter $R = 0.4$, implemented using the FastJet library [38]. To suppress contributions of fake tracks and combinatorial background (especially in the context of the larger heavy-ion background), a "hard-core" constituent selection as was done in previous STAR analyses [39] is applied, which only allows tracks (towers) with $cp_{\text{T,track}}(E_{\text{T,tower}}) \geq 2$ GeV to be clustered into jets. To enhance jet signal, only High-Tower (HT) triggered events, with at least one tower with $E_{\text{T,tower}} \geq 4$ GeV are considered. After clustering, only jets completely falling within acceptance ($|\eta_{\text{jet}}| \leq 0.6$) are kept. Jets with area, $A_{\text{jet}} < 0.3$ are rejected to further reduce the fake jet contribution.

The distributions of g, p_{T}^{D} and LeSub are fully corrected for detector effects by using iterative bayesian unfolding, implemented using the RooUnfold library [40]. The unfolding requires a response matrix between particle-level and detector-level. This is constructed using an embedding simulation which involves PYTHIA-6 STAR tune [41] events processed into detector hits using GEANT3 [42] and added to real zero-bias events from $p+p$ collision environment. To calculate $\rho(\mathbf{r})$, additional associated tracks not clustered into jets, but inside the jet cones are also used. This was done to look at the complete jet, around its hard core. Given a jet, tracks with $p_{\mathrm{T,assoc}} \geq 1$ GeV/c and $r(\mathrm{assoc}, \mathrm{jet}) \leq 0.4$ are used. The $\rho(\mathbf{r})$ is corrected using bin-by-bin factors obtained from the aforementioned embedding simulation.

5.2 Result and Discussion

Differential jet-shape as a function of $r = r(\mathrm{assoc}, \mathrm{jet})$ from the jet axis is shown in Fig. **??**. Girth (g), p_{T}^{D} and LeSub distributions are shown in Fig. **??**. Systematic uncertainties are shown as shaded grey bands. On average, lower energy jets with $15 \leq p_{\mathrm{T,jet}} < 20$ GeV/c have higher g, lower $LeSub$ and more energy away from jet-axis than jets with $p_{\mathrm{T,jet}} \geq 20$ GeV/c.

The results are compared to PYTHIA-6 (STAR) [41] and PYTHIA-8 Detroit underlying event tune [43]. All measurements show a good agreement with PYTHIA-6, while PYTHIA-8 is shown to underestimate jets with higher $LeSub$ and lower g values. $\rho(\mathbf{r})$ from PYTHIA-8 underestimates the fraction of jet momentum closer to the jet axis. Figures **??** and **??** show STAR data compared to PYTHIA-8 (Detroit) with (a) all hard scatterings, (b) only $qq \rightarrow qq$ hard scatterings (quark jets), and (c) only $gg \rightarrow gg$ hard scatterings (gluon jets).

Since gluon jets have softer, more spread-out radiation pattern on average than quark jets [44], they are likely to have lower p_{T}^{D}, lower LeSub, higher g with more momentum ($\rho(\mathbf{r})$) away from the jet-axis. As even quark-jets from PYTHIA-8 (Detroit) show softer fragmentation on average than the STAR data, it is likely that PYTHIA-8 (Detroit) underestimates hard fragmentation of partons.

5.3 Conclusions

First measurements of jet-shape observables g, p_{T}^{D}, LeSub and $\rho(\mathbf{r})$ from STAR using hard-core jets $p+p$ collisions at $\sqrt{s} = 200$ GeV are presented, setting the baseline for heavy-ion collisions to measure the medium-modification at RHIC. With the hard-core jet definition and HT trigger requirement, the sample of jets used here is biased towards hard-fragmented jets. The results show good agreement with PYTHIA-6 (STAR). PYTHIA-8 (Detroit) is shown to underestimate harder-fragmented jets, and needs further tuning of PYTHIA-8's parton shower/hadronization parameters to explain STAR hard-core jets.

Chapter 6

Conclusions & Future Work

I had a really great time in Chapters **??** & **??** & **??**, but I'm quite wearyyyyy.

Bibliography

[1] F. Takahashi et al. Review of Particle Physics. *Int. J. Mod. Phys. A*, 41:2630011, 2026.

[2] H. Agakishiev et al. Event-plane-dependent dihadron correlations with harmonic v_n subtraction in Au + Au collisions at $\sqrt{s_{NN}} = 200$ GeV. *Phys. Rev.*, C89(4):041901, 2014.

[3] A. M. Poskanzer and S.A. Voloshin. Methods for analyzing anisotropic flow in relativistic nuclear collisions. *Phys.Rev.*, C58:1671–1678, 1998.

[4] S. Acharya et al. Jet-hadron correlations measured relative to the second order event plane in Pb-Pb collisions at $\sqrt{s_{\mathrm{NN}}} = 2.76\,\mathrm{TeV}$. *Phys. Rev. C*, 101:064901, Jun 2020.

[5] W. M. Castilho, W. Qian, Y. Hama, and T. Kodama. Event-plane dependent di-hadron correlations with harmonic v_n subtraction in a hydrodynamic model. *Physics Letters B*, 777:369–373, 2018.

[6] J. Adams et al. Azimuthal anisotropy in Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV. *Physical Review C*, 72(1):014904, Jul 2005.

[7] J. Barrette et al. Proton and pion production relative to the reaction plane in Au + Au collisions at AGS energies. *Phys. Rev.*, C56:3254–3264, 1997.

[8] J. Barrette et al. Energy and charged particle flow in 10.8A GeV/c Au+Au collisions . *Phys. Rev. C*, 55:1420–1430, Mar 1997.

[9] S. A. Voloshin, A. M. Poskanzer, and R. Snellings. Collective phenomena in non-central nuclear collisions. *Landolt-Bornstein*, 23:293–333, 2010.

[10] Q. Wang. *Charge Multiplicity Asymmetry Correlation Study Searching for Local Parity Violation at RHIC for STAR collaboration.* PhD thesis, Kansas U., New York, 2012.

[11] J. Bielcikova et al. Elliptic flow contribution to two particle correlations at different orientations to the reaction plane. *Phys.Rev.*, C69:021901, 2004.

[12] S. Voloshin and Y. Zhang. Flow study in relativistic nuclear collisions by Fourier expansion of Azimuthal particle distributions. *Z.Phys.*, C70:665–672, 1996.

[13] E. Abbas et al. Performance of the ALICE VZERO system. *JINST*, 8:P10016, 2013.

[14] J. Barrette et al. Energy and charged particle flow in a 10.8A GeV/c Au + Au collisions . *Phys.Rev.*, C55:1420–1430, 1997.

[15] K. H. Ackermann et al. STAR detector overview. *Nucl. Instrum. Meth.*, A499:624–632,

2003.

[16] M. Beddo et al. The STAR Barrel Electromagnetic Calorimeter. *Nucl. Instrum. Meth.*, A499:725–739, 2003.

[17] L. Adamczyk et al. Measurements of jet quenching with semi-inclusive hadron+jet distributions in Au + Au collisions at $\sqrt{s_{NN}} = 200$ gev. *Phys. Rev. C*, 96:024905, Aug 2017.

[18] M. Cacciari, G. P. Salam, and G. Soyez. The Anti-k(t) jet clustering algorithm. *JHEP*, 04:063, 2008.

[19] M. Cacciari, G. P. Salam, and G. Soyez. FastJet User Manual. *Eur.Phys.J.*, C72:1896, 2012.

[20] M. Cacciari, G. P. Salam, and G. Soyez. The Catchment Area of Jets. *JHEP*, 04:005, 2008.

[21] L. Adamczyk et al. Dijet imbalance measurements in Au + Au and *pp* collisions at $\sqrt{s_{\mathrm{NN}}} = 200$ GeV at star. *Phys. Rev. Lett.*, 119:062301, Aug 2017.

[22] N. Sharma et al. Background subtraction methods for precision measurements of di-hadron and jet-hadron correlations in heavy ion collisions. *Phys. Rev.*, C93(4):044915, 2016.

[23] G. Aad et al. Measurement of event-plane correlations in $\sqrt{s_{NN}} = 2.76$ TeV lead-lead collisions with the ATLAS detector. *Phys. Rev.*, C90(2):024905, 2014.

[24] C. Nattrass and T. Todoroki. Event plane dependence of the flow modulated background in dihadron and jet-hadron correlations in heavy ion collisions. *Physical Review C*, 97(5), May 2018.

[25] C. Nattrass. Re-examining the iconic dihadron correlation measurement demonstrating jet quenching. *Physical Review C*, 97(3), Mar 2018.

[26] C. Nattrass et al. Disappearance of the Mach Cone in heavy ion collisions. *Phys. Rev.*, C94(1):011901, 2016.

[27] Korinna Zapp, Gunnar Ingelman, Johan Rathsman, Johanna Stachel, and Urs Achim Wiedemann. A monte carlo model for 'jet quenching'. *The European Physical Journal C*, 60(4), mar 2009.

[28] Korinna C. Zapp. JEWEL 2.0.0: directions for use. *Eur. Phys. J.*, C74(2):2762, 2014.

[29] J. G. Milhano and K. C. Zapp. Origins of the di-jet asymmetry in heavy ion collisions. *Eur. Phys. J.*, C76(5):288, 2016.

[30] K. C. Zapp. Geometrical aspects of jet quenching in JEWEL. *Phys.Lett.*, B735:157–163, 2014.

[31] J. Noronha-Hostler. Solving the $R_{AA} \bigotimes v_2$ puzzle. *Journal of Physics: Conference*

Series, 736(1), August 2016.

[32] J. Noronha-Hostler et al. Event-by-event Hydrodynamics + Jet energy loss: A solution to the $R_{AA} \bigotimes v_2$ puzzle. *Phys. Rev. Lett.*, 116:252301, Jun 2016.

[33] Andrew J. Larkoski, Jesse Thaler, and Wouter J. Waalewijn. Gaining (mutual) information about quark/gluon discrimination. *Journal of High Energy Physics*, 2014(11), nov 2014.

[34] S. Chatrchyan et al. Modification of jet shapes in PbPb collisions at $\sqrt{s_{\rm NN}} = 2.76$TeV. *Physics Letters B*, 730:243–263, 2014.

[35] S. Acharya et al. Medium modification of the shape of small-radius jets in central Pb-Pb collisions at 2.76 TeV. *Journal of High Energy Physics*, 2018(10), oct 2018.

[36] M. Anderson et al. The STAR time projection chamber: a unique tool for studying high multiplicity events at RHIC. *Nuc. Inst. and Methods in Physics Research*, 499(2-3):659–678, mar 2003.

[37] M. Beddo et al. The STAR barrel electromagnetic calorimeter. *Nuc. Inst. and Methods in Physics Research*, 499(2):725–739, 2003. The Relativistic Heavy Ion Collider Project: RHIC and its Detectors.

[38] Matteo Cacciari, Gavin P. Salam, and Gregory Soyez. FastJet user manual. *The European Physical Journal C*, 72(3), mar 2012.

[39] M. S. Abdallah et al. Differential measurements of jet substructure and partonic energy loss in Au+Au collisions at $\sqrt{s_{\rm NN}} = 200$ GeV. *Phys. Rev. C*, 105:044906, Apr 2022.

[40] Lydia Brenner et al. Comparison of unfolding methods using RooFitUnfold. *International Journal of Modern Physics A*, 35(24):2050145, 2020.

[41] J. Kevin Adkins. PhD thesis, University of Kentucky, 2019.

[42] R. Brun et al. GEANT3. 9 1987.

[43] Manny Rosales Aguilar et al. Pythia8 underlying event tune for RHIC energies. *Phys. Rev. D*, 105:016011, Jan 2022.

[44] Jason Gallicchio and Matthew D. Schwartz. Quark and gluon tagging at the LHC. *Physical Review Letters*, 107(17), oct 2011.